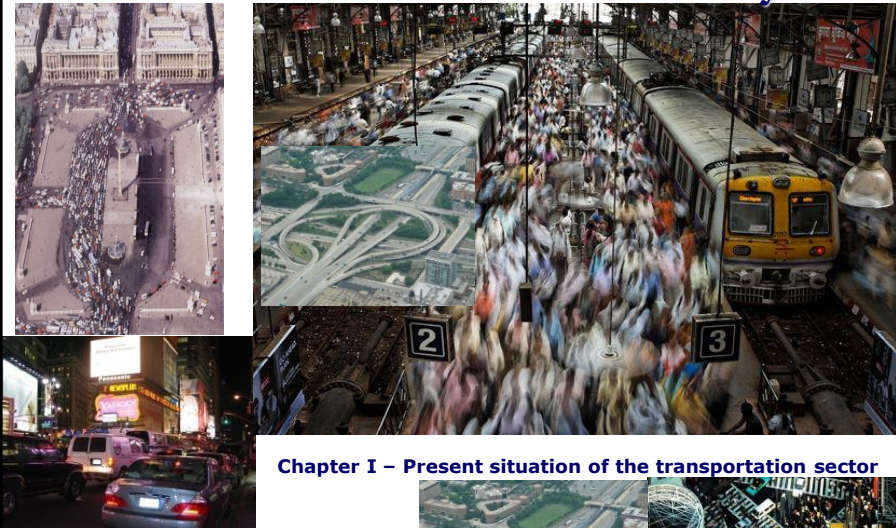




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Sustainable Mobility



Chapter I – Present situation of the transportation sector

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2022/2023 – 2nd Semester

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REFERENCES

- TERM 2016, TERM2017, TERM 2018, TERM2019, TERM2020, European Environment Agency.
- “EU Energy and Transport in Figures – Statistical Pocketbook 2021”, European Commission, 2021.
- Statistics from ACAP, INE, Instituto de Seguros de Portugal
- Deloitte (2021) Macrotendências para o Cluster da Indústria Automóvel, MOBINOV.
- EU policy documents:
 - Roadmap to a single European transport area —Towards a competitive and resource efficient transport system (referred to as the 2011 Transport White Paper) (2011)
 - Smart and sustainable mobility strategy (2020)

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CLASS#1

AN INSPIRATION TO START...



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Evolution of Mobility

Easter morning 1900: 5th Ave, New York City. Spot the automobile.



Source: US National Archives.

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Evolution of Mobility

Easter morning 1913: 5th Ave, New York City.
Spot the horse.



Source: George Grantham Bain Collection.

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Evolution of Mobility



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Why is important the module “Sustainable Mobility”?...

- 56.2% of the global population now lives in cities $\Rightarrow$ 2/3 of global population will live in cities by 2050.
- Instability of oil price /fuel consumption
- Impacts at $\neq$ levels: traffic congestion & road safety & GHGs emissions & pollutant emissions & noise
- Importance of alternative fuels / propulsion modes

Energy Consumption: The Burguer Comparison

1 BURGUER = 590 kcal or 2.5 MJ



(*) The diagram of a motor vehicle is the group of components that deliver power to the driving wheels. This includes the transmission, the wheels and the axles.
Ref: EEA, Explaining road transport emissions, 2016.



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Motivation

- Europe: ~33% of **energy consumption** from transportation sector
- Road: ~73% of transportation-related GHG **emissions**
- Vehicle emissions → important source for **air pollution** (NOx, PM) → 400,000 premature deaths/yr due to air pollution in Europe
- Road transportation: dominant source of **noise** in Europe (~90% of total noise emissions)
- Road **crashes**: globally, >1 million people die every yr
- Conventional fuels are expected to remain predominant over the next decades



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READY TO DIVE INTO STATISTICS?

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WORLD TRANSPORTATION

- > increase: road and air modes
- There are more than 1000 million of road vehicles in the World. 2050 $\Rightarrow$ projection of more than 2000 million
- World average motorization rate in 2010: 148 veh/1000 people.
- Some numbers $\rightarrow$

		Infrastructure and Vehicles				
		EU-27	USA	Japan	China	Russia
		Transport infrastructure				
		2019	2019	2018	2019	2019
Road network (paved)	1000 km	4409,8	4 676	1 008	4 699	1 189
Motorway network	1000 km	74,0	92,5 (1)	9,4 (2)	149,6	57,0 (3)
Railway network	1000 km	200,2	198,4 (4)	19,1	140,0	85,5
Electrified rail lines	1000 km	112,1		11,8	75,9	44,0
Navigable inland waterways	1000 km	41,9	40,2		127,0	102,0
Oil Pipelines	1000 km	31,7	362,1		126,6 (5)	70,0
		Vehicle stock				
		2019	2019	2019	2019	2019
Passenger cars stock	million	247,4	251 (6)	61,8 (7)	137,0	48,4
Motorization	cars / 1000 persons	553	764	490	98	335
Commercial freight vehicles	million	35,61	13,09	14,19 (8)	27,83	6,54
		Transport safety				
		2019	2019	2019	2018	2019
Road fatalities	number per million inhabitants	22 763	36 096	3 920	63 194	16 981
		51	110	31	45	118

Source: Chapter 2.5, 2.6 and 2.7, International Transport Forum (Statistics), Union Internationale des Chemins de Fer (JP, CN), national statistics (CN, JP, RU, US), estimates (in italics)

NB:

(1): Divided highways with 4 or more lanes (rural or urban interstate, freeways, expressways, arterial and collector) with full access control by the authorities.

(2): Japan: national expressways.

(3): RU: roads of federal importance

(4): USA: a sum of partly overlapping networks.

(5): China: both oil and gas pipelines

(6): USA: light duty vehicles, short wheel and long wheel base.

(7): Japan: ordinary, small and light four-wheeled vehicles.

(8): Japan: including 8.27 million light motor vehicles.

Ref: EU Transport in Figures, 2021

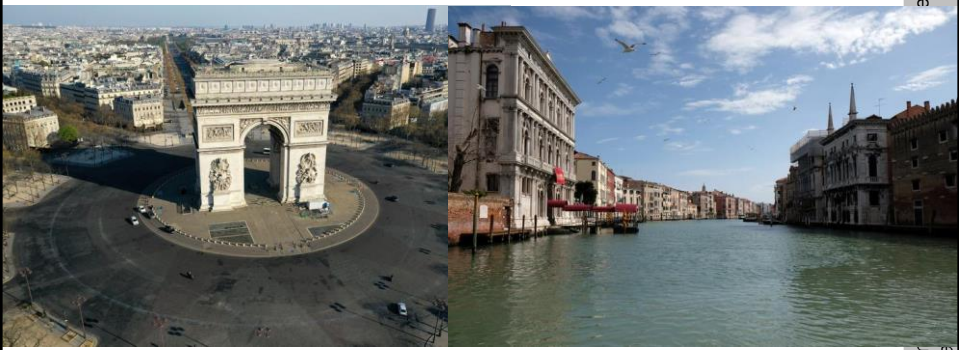
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WORLD TRANSPORTATION... until March 2020

COVID-19 and Mobility



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WORLD TRANSPORTATION... until March 2020

COVID-19 and Mobility




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WORLD TRANSPORTATION... until March 2020

COVID-19 and Mobility

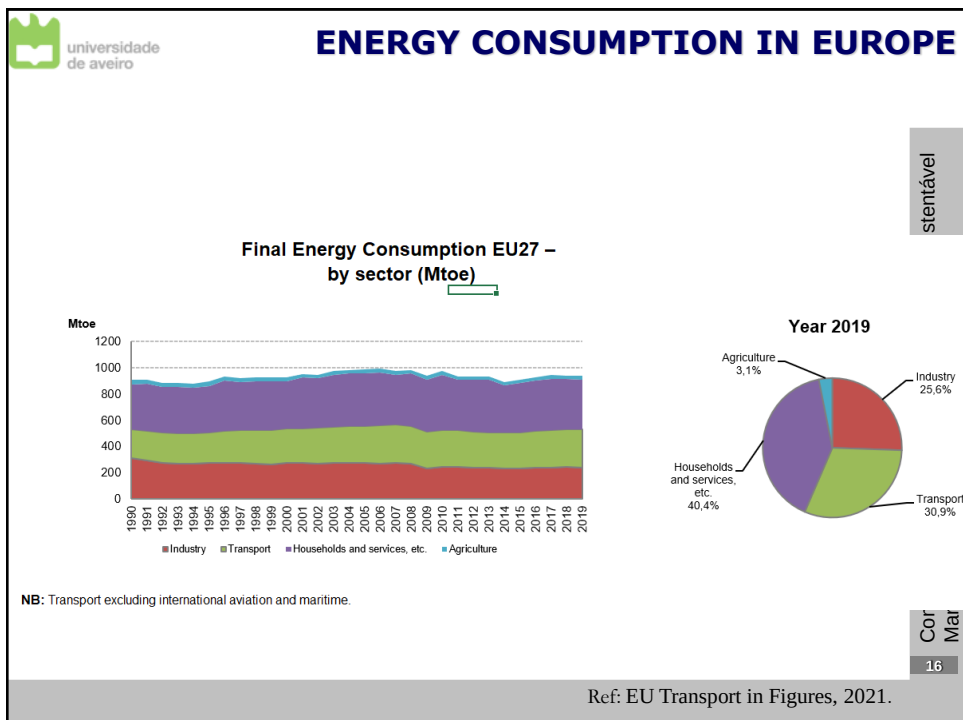
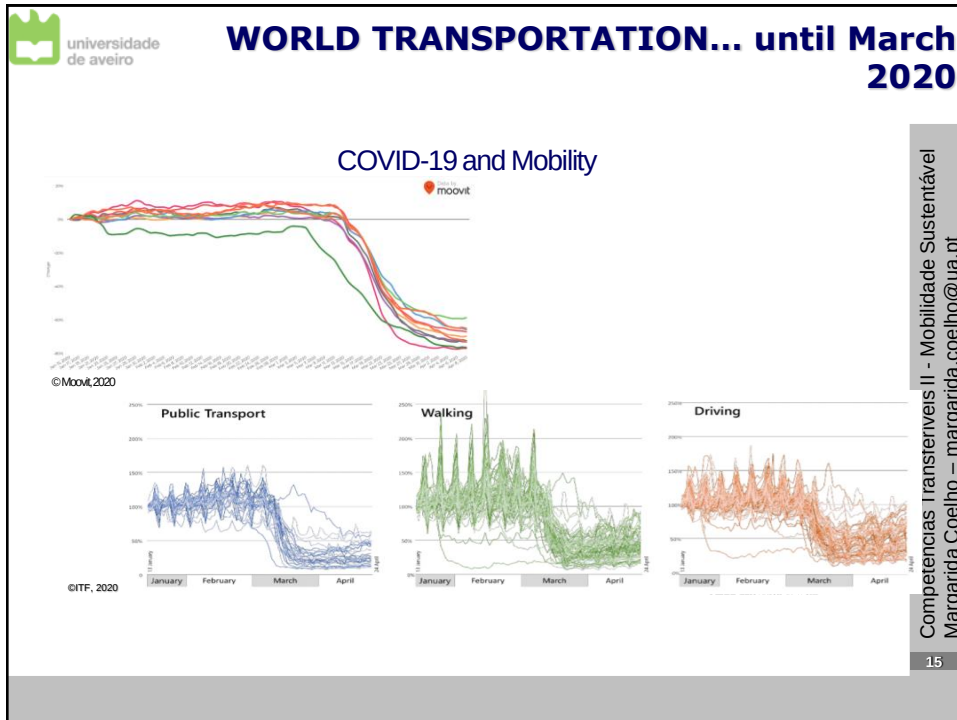


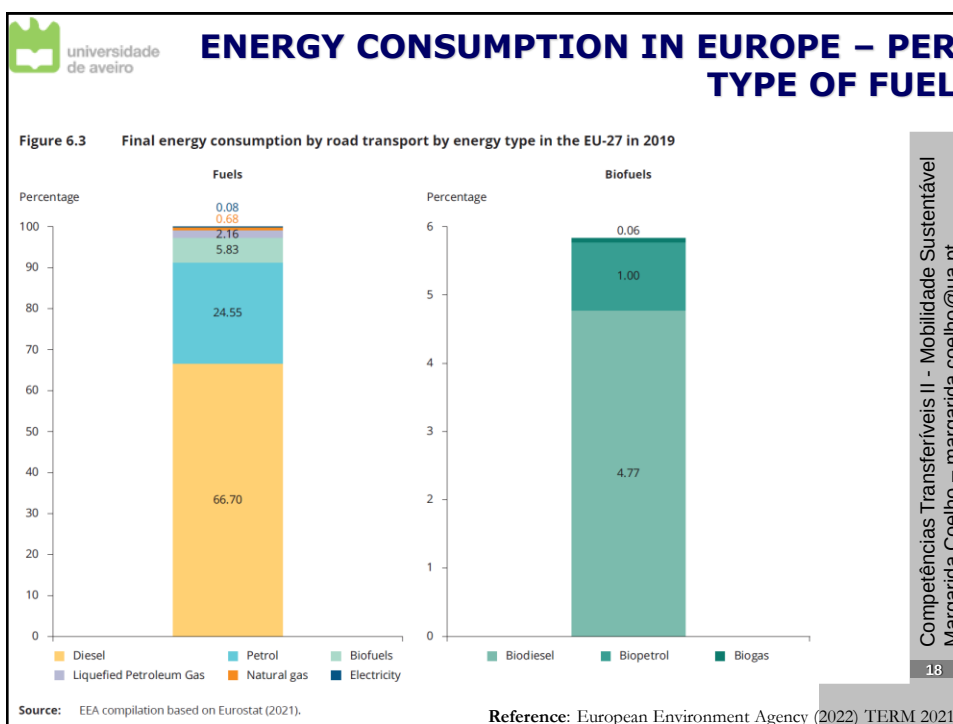
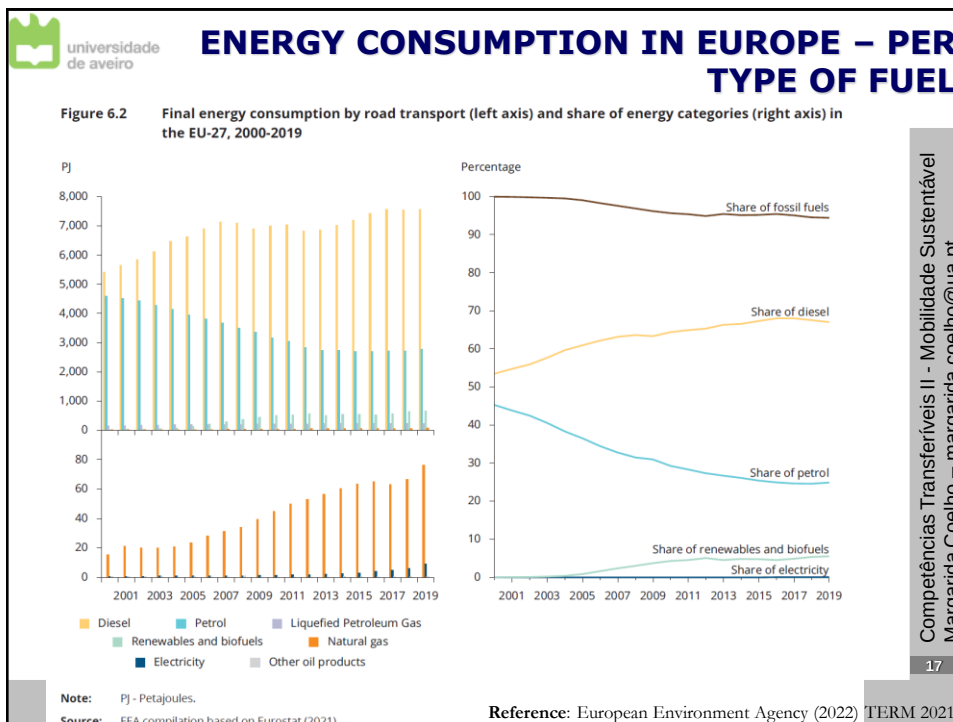

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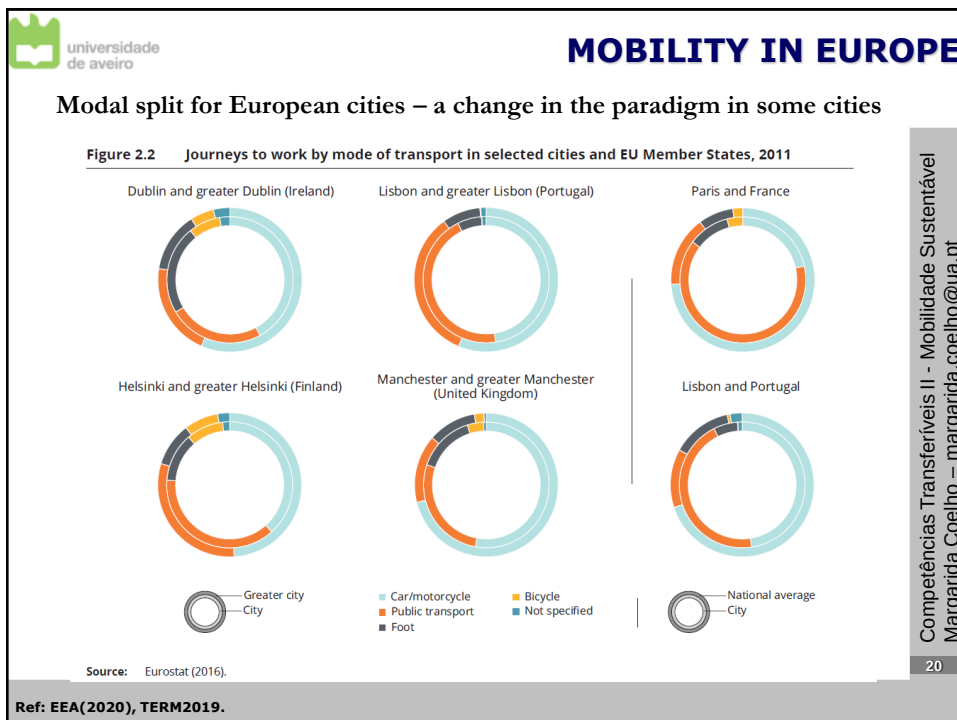
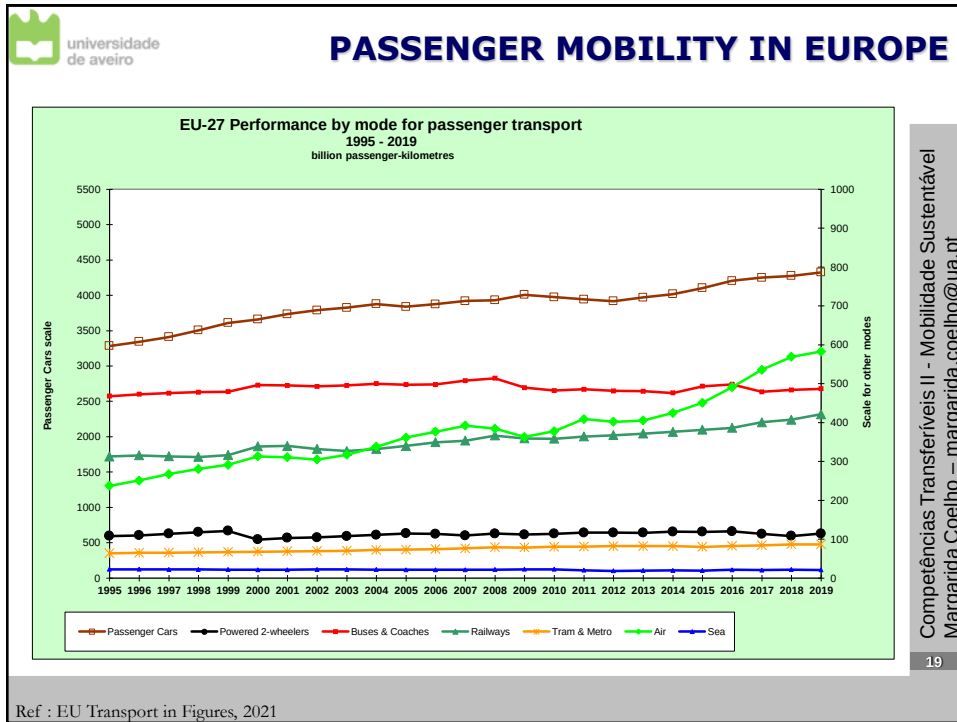
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PASSENGER MOBILITY IN EUROPE – WHAT ABOUT CYCLING?

- There are no comprehensive EU-wide data on **cycling** and other forms of non-motorised passenger transport.

⇒ On average 8% of respondents to the Eurobarometer survey mention cycling as the most important mode of travel on a typical day.

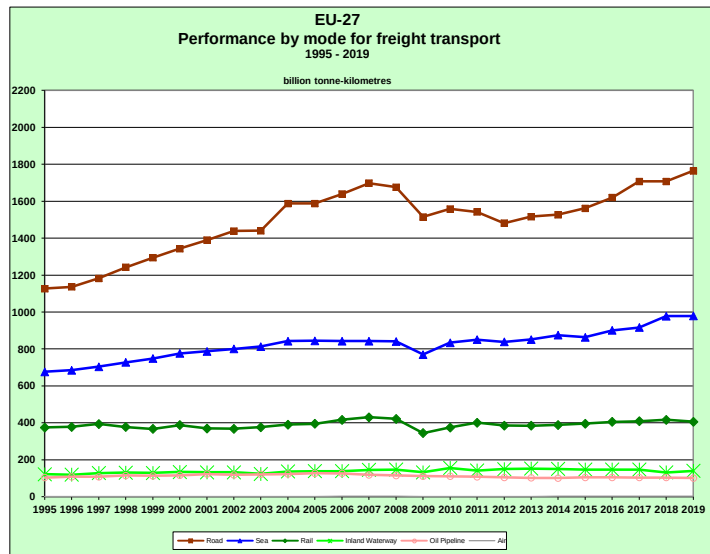
⇒ This ranges from cycling rates of 1% or less in Cyprus, Malta, and Portugal to 36% in the Netherlands. In the latter, the distance travelled by bicycle increased by 14% between 2000 and 2012, with most of this growth occurring as a result of increased popularity and the availability of electric bicycles

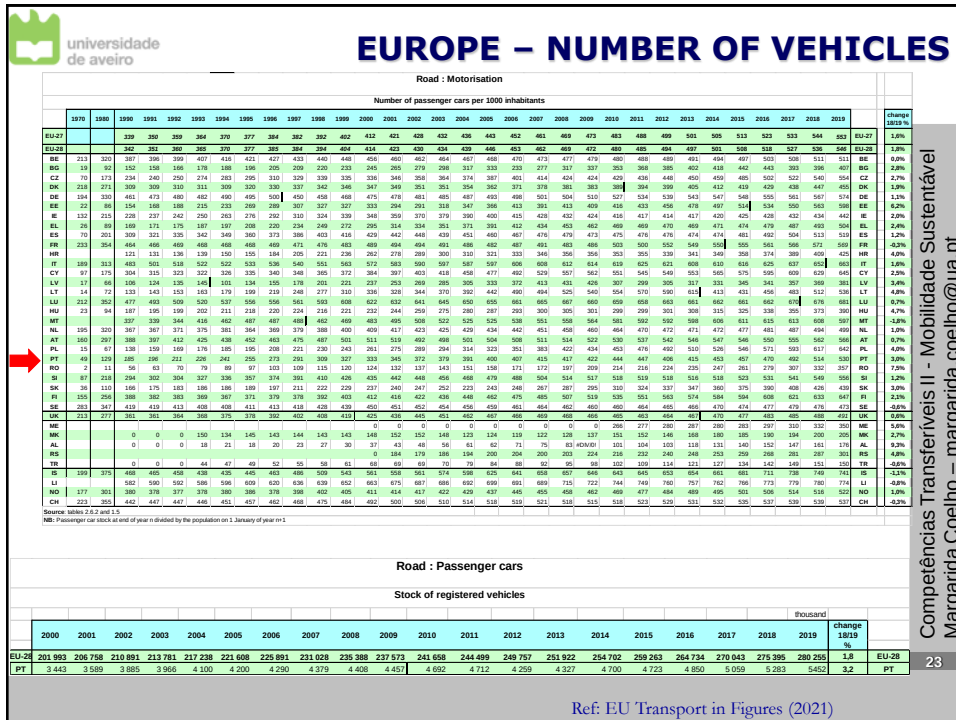
- A huge interest on cycling during the pandemic (pop-up bike lanes): new bicycle lanes created during the COVID-19 pandemic have increased cycle traffic in European cities by between 11% and 48%.

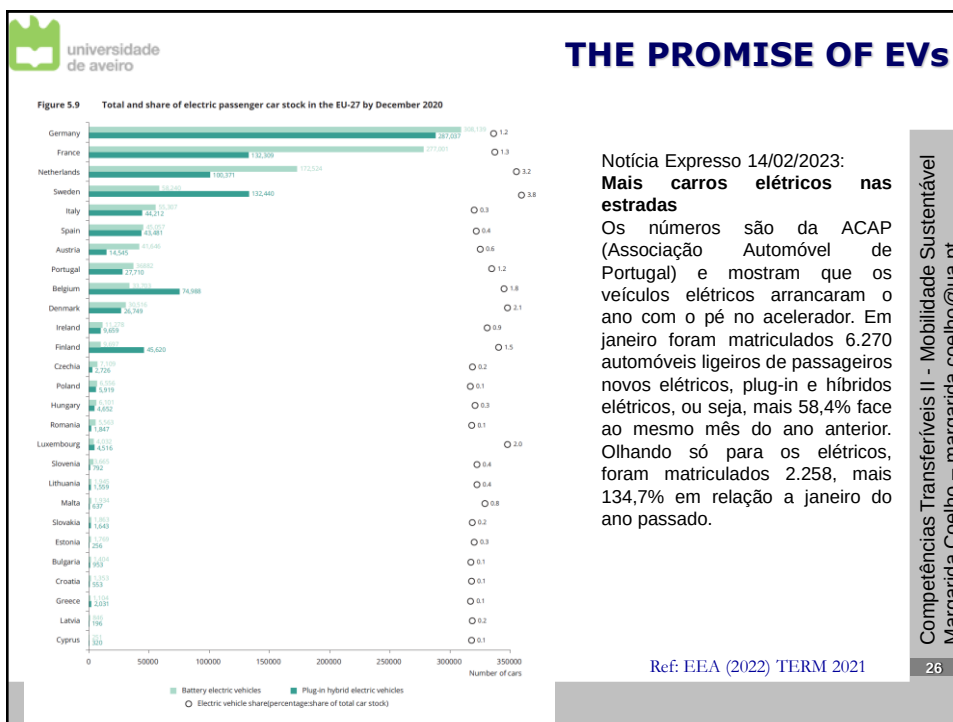
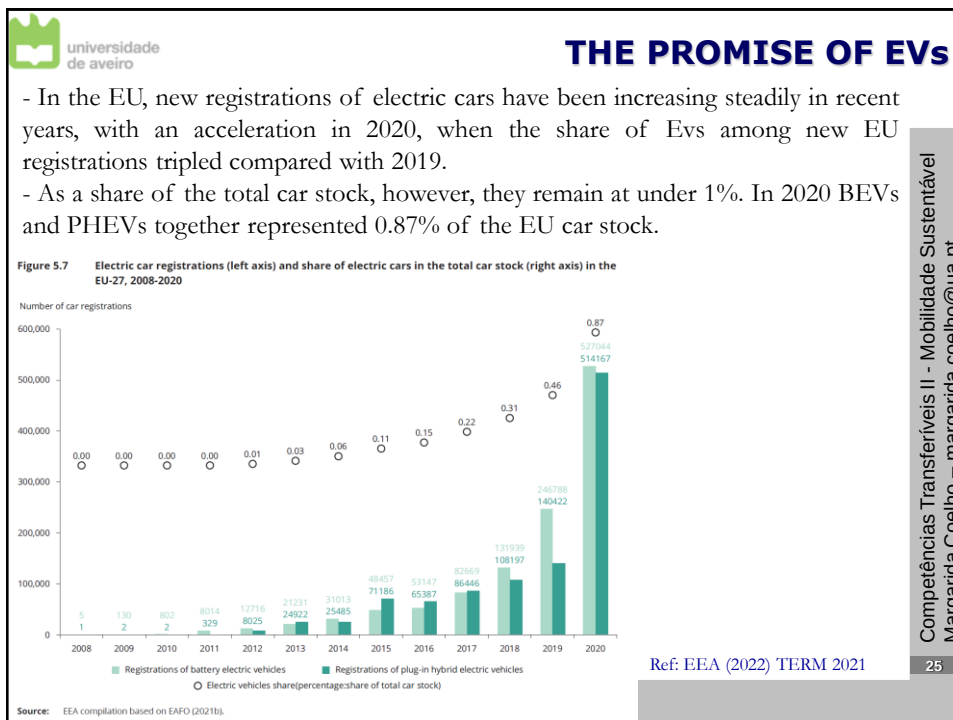
(Ref. Guardian)



FREIGHT MOBILITY IN EUROPE









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TRANSPORTATION IN PORTUGAL - ENERGY

Portugal

2019:

- **6.0 Mtep: transportation (5,7 for road transport)**
- **4.6 Mtep: industry**
- **5.8 Mtep: household & services & agriculture & fishing**

Total: ~16.4 Mtep

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Ref: EU Energy in Figures, 2021.



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MOBILITY IN PORTUGAL

- ~ 6,0 million cars in Portugal (peak in 2017)
- Car ownership rate > 500
 - 1987: ~190 cars/1000 inhab.
 - 2022: >500 cars/1000 inhab.
- Average age of cars: 12,6 yr (2017)
- ~370 000 veh enter everyday in Lisbon (+200 000 that circulate there)

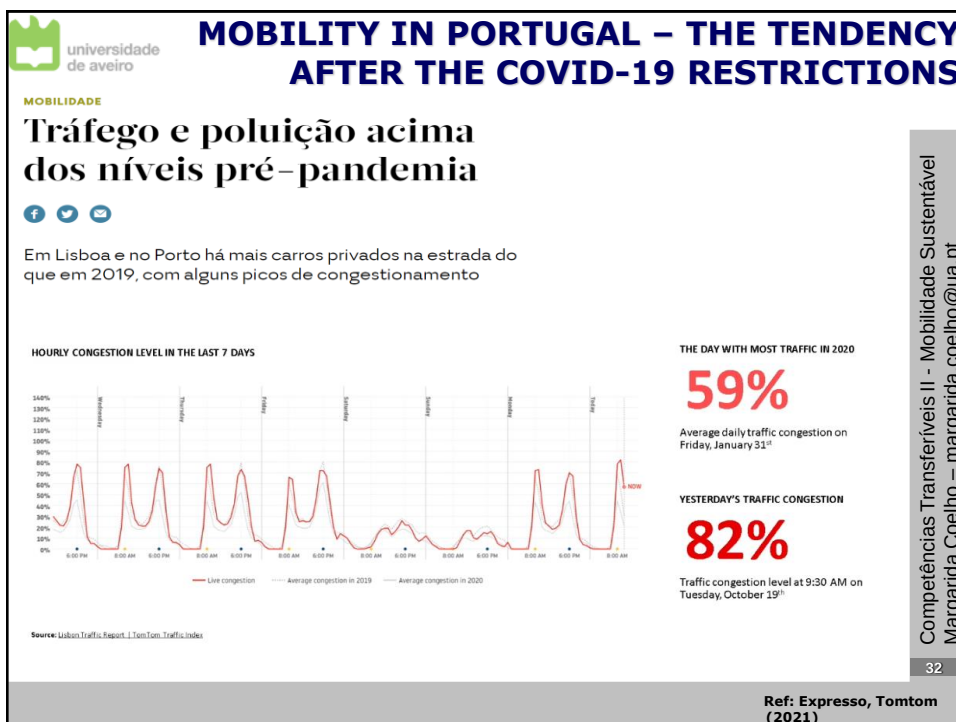
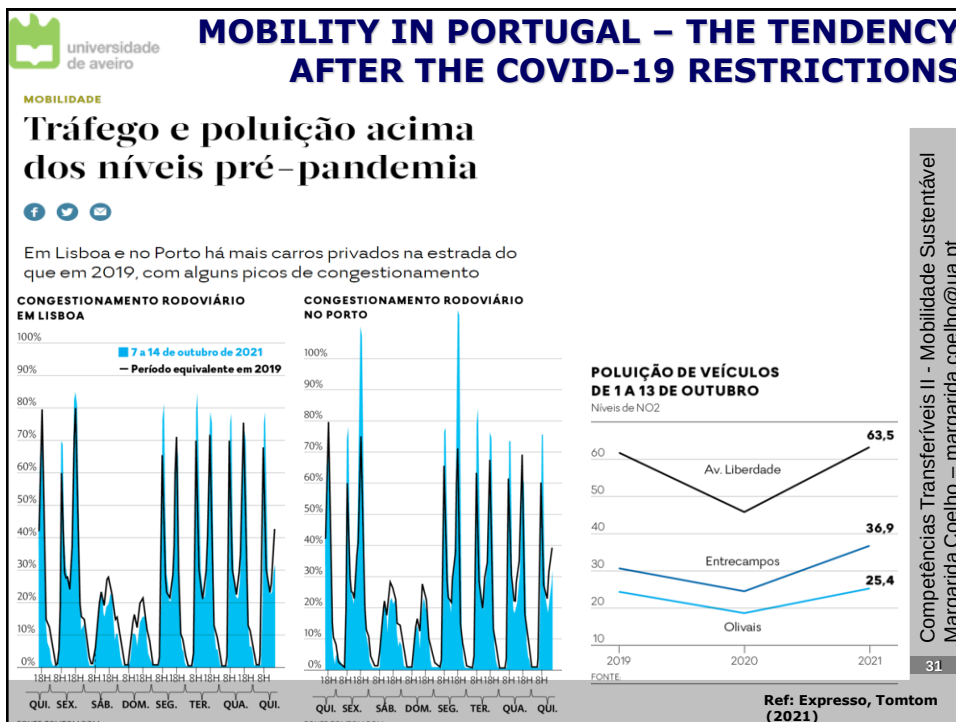


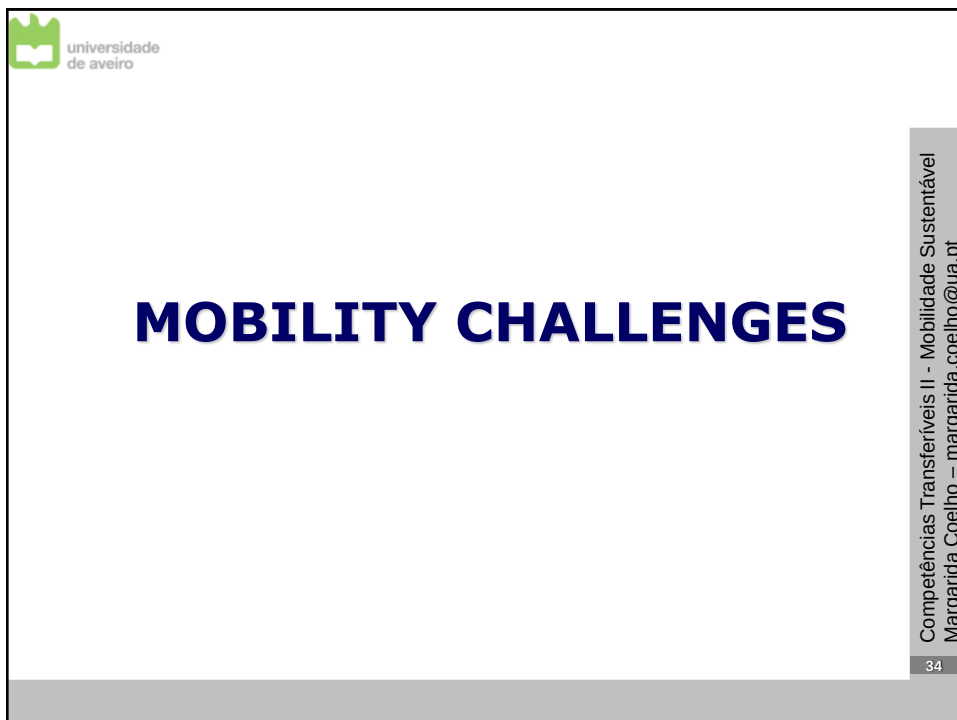
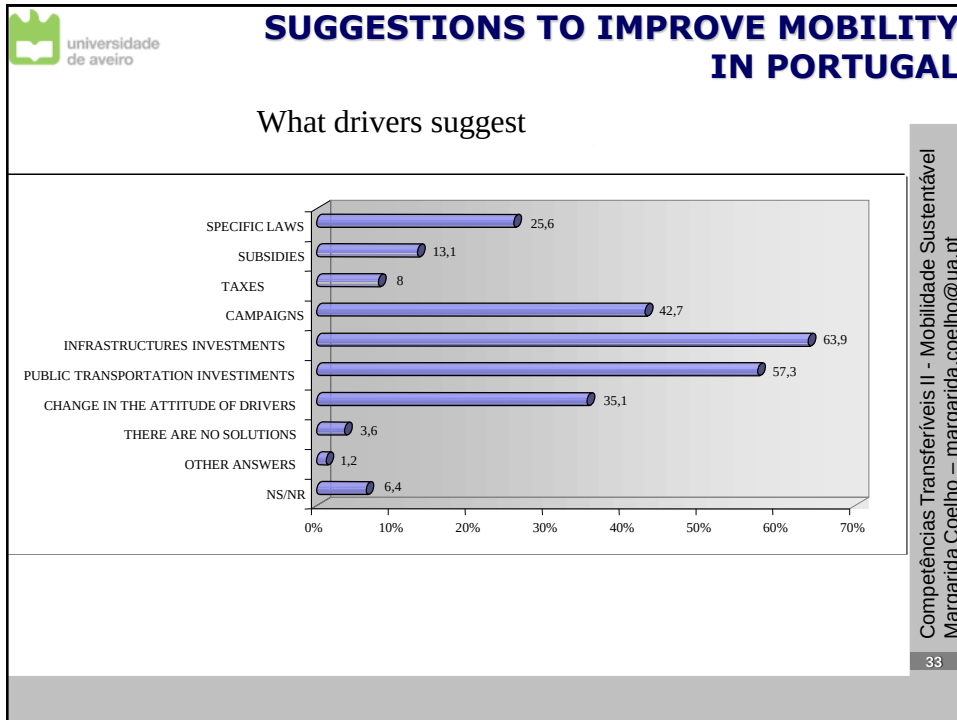
Mobility survey - results launched in August 2018:

- Car: ~69% trips at Metropolitan Area of Porto (AMP); ~59.8% trips at Metropolitan Area of Lisbon (AML)
- Active modes (walking & bicycle): 18,9% AMP; 23,5% na AML... specially for shopping (29,6% AMP and 38,9% AML)!
- Lisbon citizens use more public transportation: 15,8% AML vs. 11,1% AMP.

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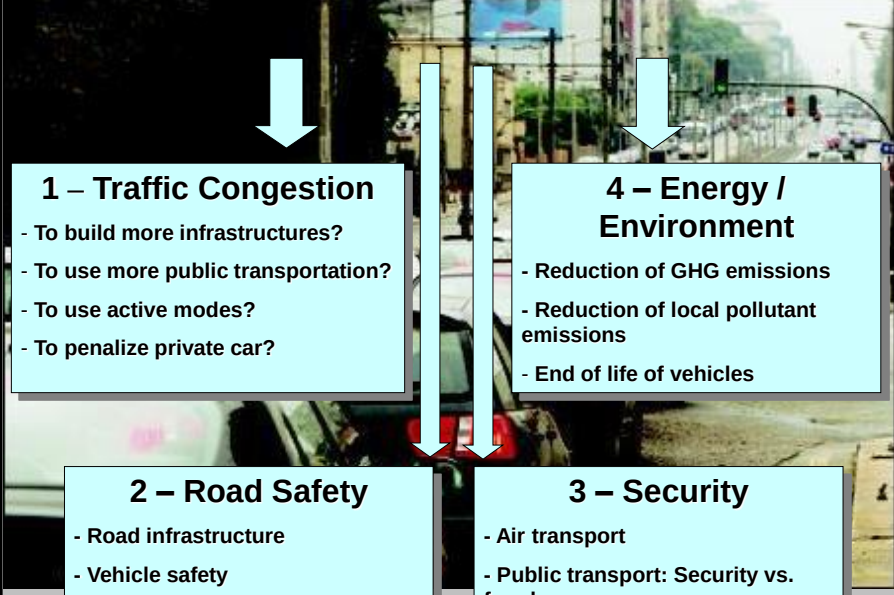
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BIGGEST CHALLENGES ON TRANSPORTATION SECTOR



1 – Traffic Congestion

- To build more infrastructures?
- To use more public transportation?
- To use active modes?
- To penalize private car?

4 – Energy / Environment

- Reduction of GHG emissions
- Reduction of local pollutant emissions
- End of life of vehicles

2 – Road Safety

- Road infrastructure
- Vehicle safety
- Human factor

3 – Security

- Air transport
- Public transport: Security vs. freedom

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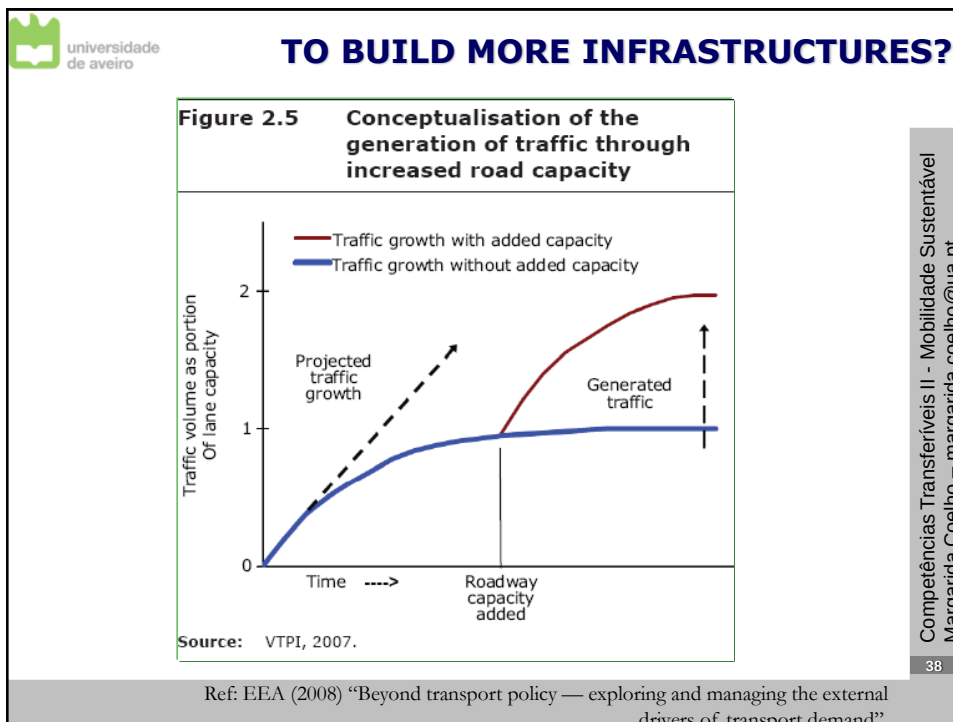
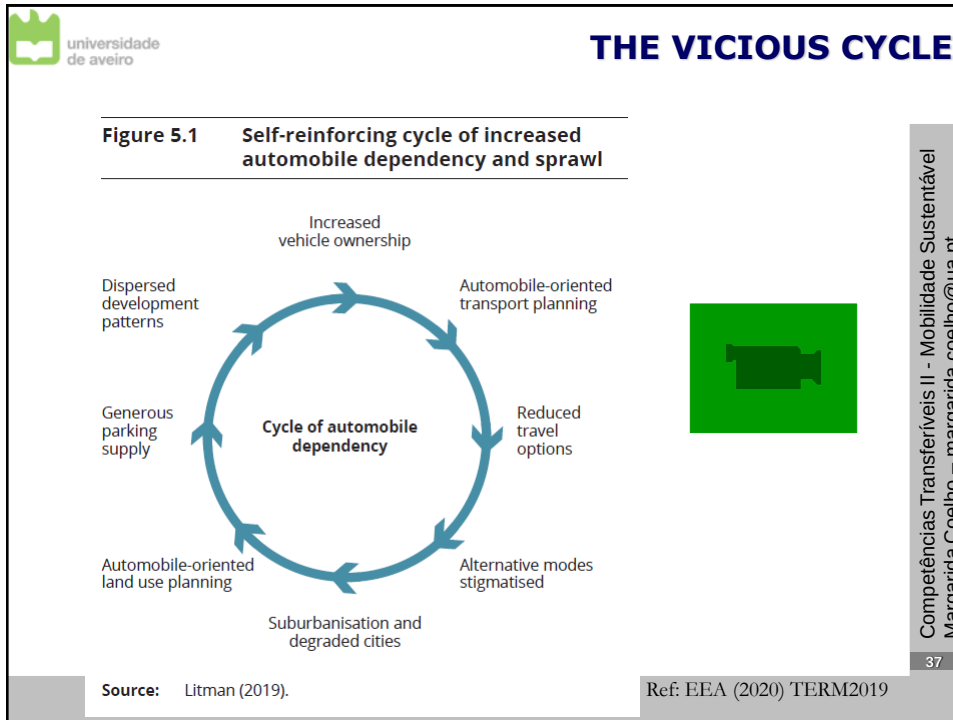
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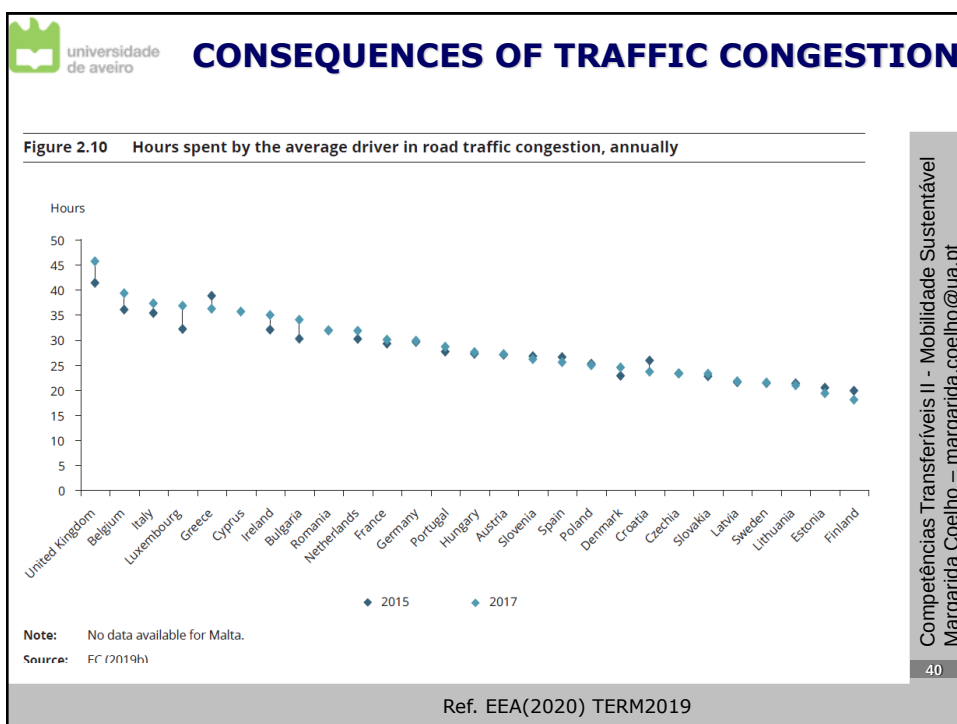
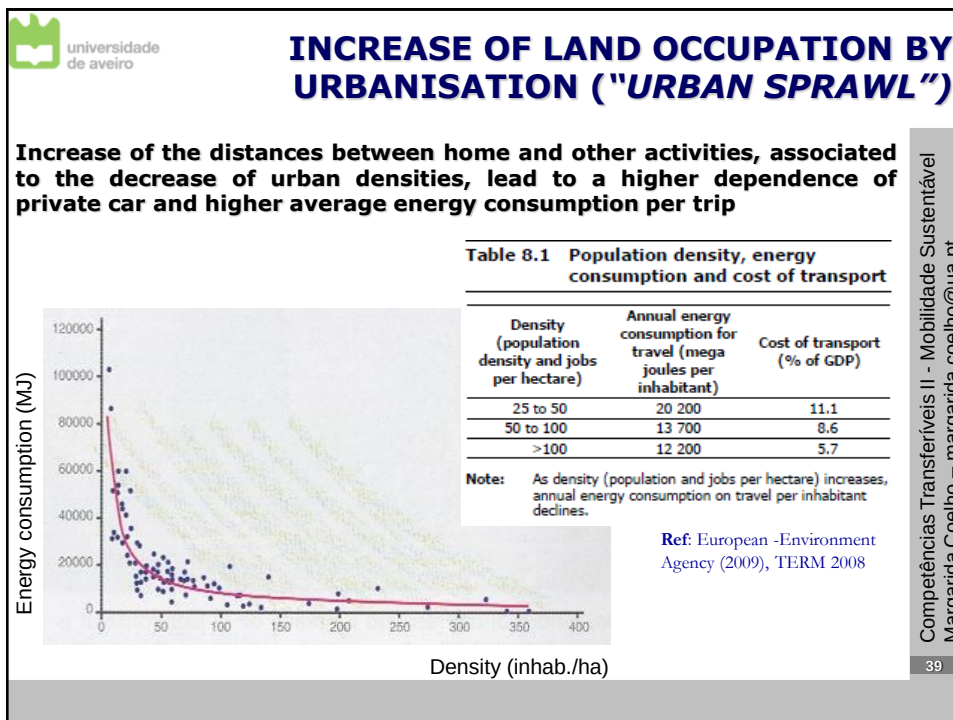
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CHALLENGE #1 – TRAFFIC CONGESTION

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CONSEQUENCES OF TRAFFIC CONGESTION: THE TOP CITIES

Table 2.2 The 15 urban areas in Europe with most hours lost in congestion (2018)

Urban area	Hours lost in congestion	Change compared with 2017 (%)
Rome (IT)	254	+16
Dublin (IE)	246	-4
Paris (FR)	237	+7
London (UK)	227	+1
Milan (IT)	226	+6
Bordeaux (FR)	223	+12
Florence (IT)	195	+3
Brussels (BE)	195	+1
Belfast (UK)	190	-10
Naples (IT)	186	-3
Warsaw (PL)	173	+7
Turin (IT)	167	-2
Edinburgh (UK)	165	-10
Montpellier (FR)	163	+9
Budapest (HU)	162	+11

Note: Urban area = the geographical scope of a city as defined by its road network density; hours lost in congestion = the total number of hours lost in congestion per commuter during peak commute periods compared with free-flowing conditions.

Source: INRIX Research (2019).

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Ref. EEA(2020) TERM2019

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CONSEQUENCES OF TRAFFIC CONGESTION



“Place de la Concorde”

- ⇒ Metropolitan Area of Paris
 - 600 millions of hours that are lost / year
- ⇒ **Consequences on economy and environment**
 - lost productivity
 - delays on the distribution of goods
 - reduction of average speeds of buses
 - > emission rates due to stop-and-go situations.
 - non efficient and illegal occupation of the urban space
 - annual loss of 500 000 million € (developed countries)
 - air pollution ~ 1-2% of GDP
 - > 30% of European population exposed to high noise levels

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NON EFFICIENT USE OF TRANSPORT MODES



75 people can travel in ~ 60 cars...

...or by 1 bus!

Considering the load capacity in vehicles...



NON EFFICIENT USE OF TRANSPORT MODES

Comparação do transporte de 220 pessoas em distintos meios de transporte

- Consumo de Energia
- Emissão de gases
- Consumo de Espaço urbano

174 Automóveis



5.500 kWh (100%)

8.248 gr (100%)



3 Autocarros



716 kWh (13%)

945 gr (11%)



1 Eléctrico ou ML

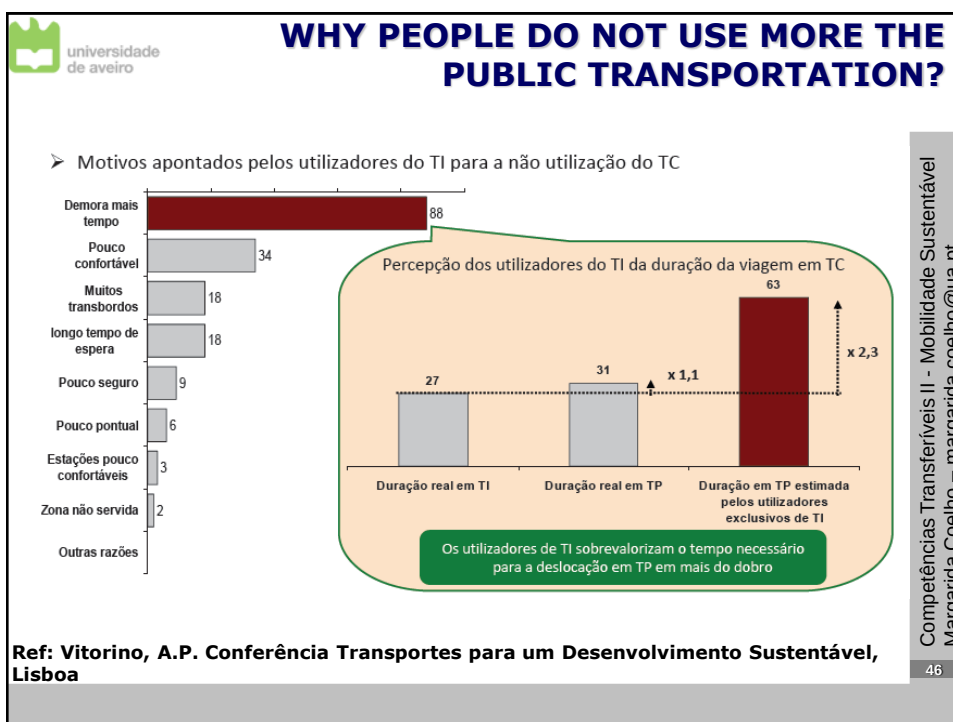
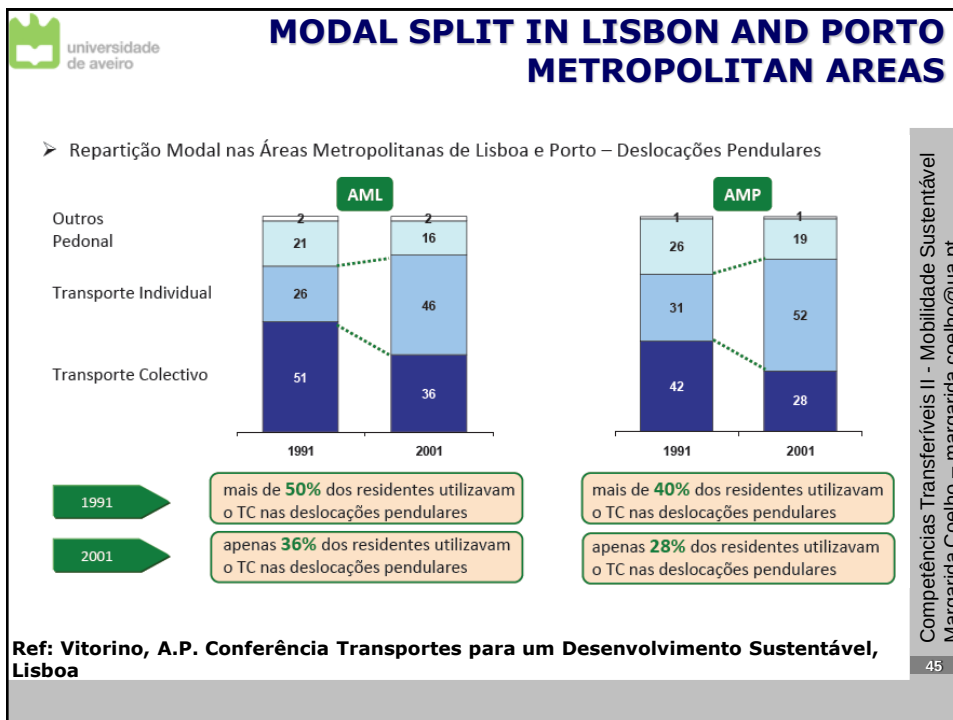


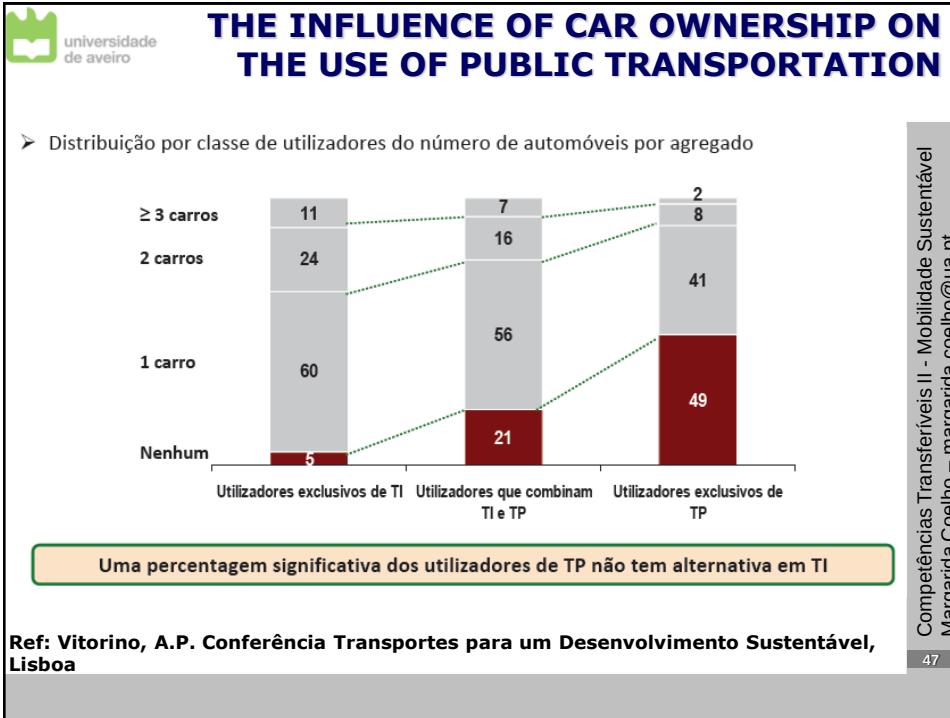
360 kWh (7%)

0 gr



Ref: Vitorino, A.P. Conferência Transportes para um Desenvolvimento Sustentável, Lisboa

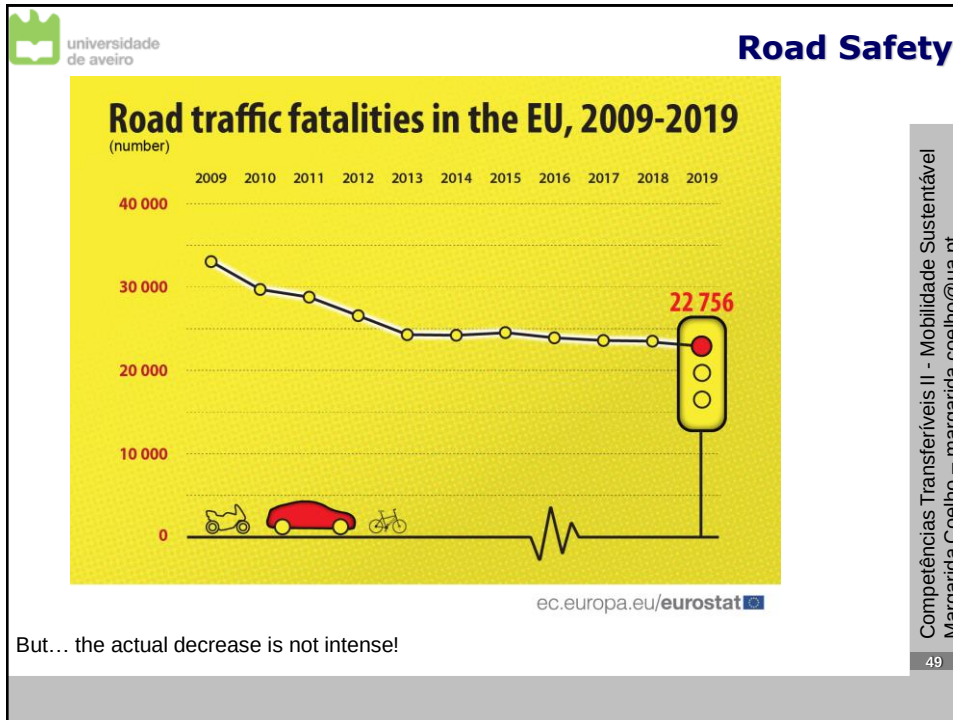




CHALLENGE #2 – ROAD SAFETY

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Road Safety

Road fatalities - country rankings					
Fatalities 2019					
per million inhabitants		per 10 billion pkm		per million passenger cars	
SE	22	SE	19	SE	45
IE	28	IE	24	LU	52
MT	32	LU	27	MT	53
NL	34	FI	31	FI	60
DK	34	DK	32	DE	64
LU	35	DE	33	IE	65
DE	37	EE	36	EE	67
ES	37	SI	36	NL	68
FI	38	NL	39	ES	72
EE	39	FR	41	DK	76
AT	47	IT	41	IT	81
FR	48	ES	50	AT	83
SI	49	EU-27	51	FR	85
SK	50	AT	51	SI	88
EU-27	51	MT	56	CY	93
IT	53	LT	59	EU-27	93
BE	56	BE	59	CZ	106
CZ	58	EL	62	BE	110
CY	59	PT	69	SK	115
HU	62	CZ	72	PL	122
EL	64	CY	73	LT	127
LT	67	LV	82	PT	128
PT	67	HU	88	EL	129
LV	69	SK	90	HU	162
HR	73	BG	105	HR	175
PL	77	HR	116	LV	184
BG	90	PL	117	BG	224
RO	96	RO	162	RO	279

Ref: EU Transport in Figures (2021)

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Road Safety

Road Fatalities by Type of User

	Year	Total	Drivers	Passengers	Cyclists	Moped / Motorbike riders and passengers	Pedestrians	Vulnerable road users as % of total	Pedestrians as % of total	
BE	2019	646	276	76	95	103	92	44.9%	14.2%	BE
BG	2019	638	299	141	27	7	154	29.9%	24.5%	BG
CZ	2019	618	265	103	53	86	111	40.5%	18.0%	CZ
DK	2019	199	78	20	31	40	30	50.8%	15.1%	DK
DE	2019	3 046	1 252	323	445	605	421	48.3%	13.8%	DE
EE	2019	52	23	10	2	4	13	36.5%	25.0%	EE
IE	2016	182	77	38	10	22	35	36.8%	19.2%	IE
EL	2019	688	216	58	22	247	145	60.2%	21.1%	EL
ES	2019	1 755	615	213	80	466	381	52.8%	21.7%	ES
FR	2019	3 237	1 404	421	187	749	476	43.6%	14.7%	FR
HR	2019	297	116	49	16	55	61	44.4%	20.5%	HR
IT	2019	3 173	1 215	385	253	786	534	49.6%	16.8%	IT
CY	2019	52	16	5	1	16	13	57.7%	25.0%	CY
LV	2019	132	60	21	9	2	40	38.6%	30.3%	LV
LT	2015	242	69	52	22	16	81	49.2%	33.5%	LT
LU	2019	22	12	4	0	4	2	27.3%	9.1%	LU
HU	2019	602	191	130	63	74	144	46.7%	23.9%	HU
MT	2018	18	4	3	1	8	2	61.1%	11.1%	MT
NL	2019	586	226	-	148	80	49	47.3%	8.4%	NL
AT	2019	416	172	51	33	91	69	46.4%	16.6%	AT
PL	2019	2 909	949	527	258	382	793	49.3%	27.3%	PL
PT	2019	688	224	134	27	163	140	48.0%	20.3%	PT
RO	2019	1 864	485	383	198	69	729	53.4%	39.1%	RO
SI	2019	102	35	20	9	23	15	46.1%	14.7%	SI
SK	2019	270	107	64	17	2	80	36.7%	29.6%	SK
FI	2019	211	115	30	23	28	15	31.3%	7.1%	FI
SE	2019	221	94	40	17	34	27	35.3%	12.2%	SE
UK	2018	1 839	642	264	100	361	472	50.7%	26.7%	UK

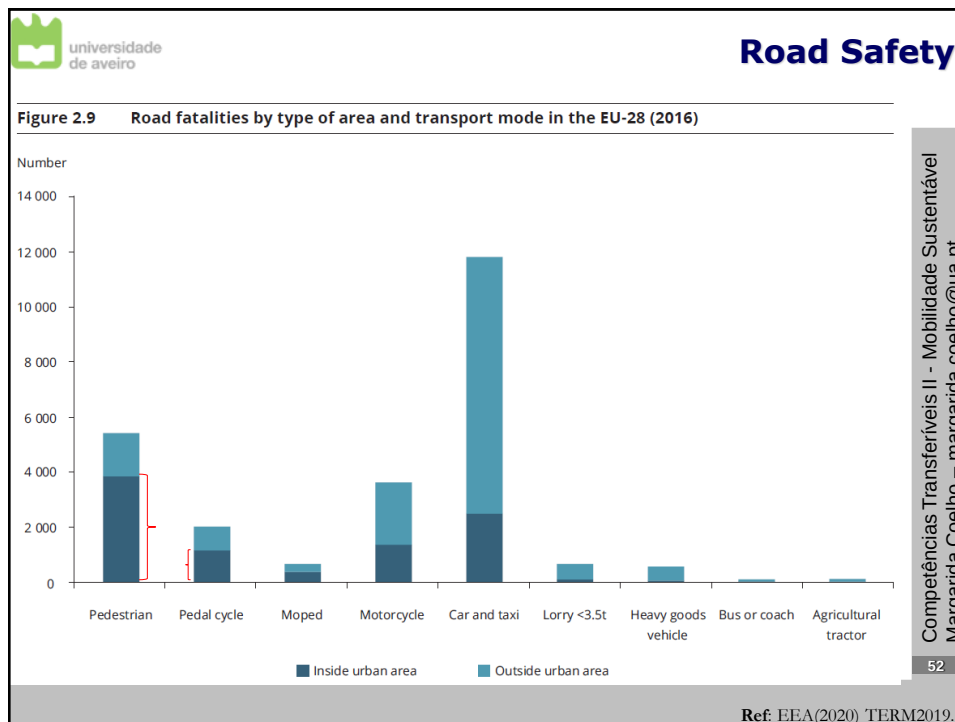
Source: CARE database (DG Mobility and Transport);

Notes:
Persons deceased within 30 days of their accident. Totals include the victims labelled as "Unknown". Vulnerable road users comprise pedestrians, cyclists, moped and motorbike riders. NL - 2019 data on passengers not available.

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Ref: EU Transport in Figures (2021).

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Road Safety

2,100 LIVES COULD BE SAVED EACH YEAR IF THE AVERAGE SPEED DROPPED BY ONLY 1 KM/H ON ALL ROADS ACROSS THE EU

ISA = **20% FEWER DEATHS**

THE MANDATORY INSTALLATION OF AN OVERRIDABLE INTELLIGENT SPEED ASSISTANCE (ISA) SYSTEM ON ALL NEW VEHICLES COULD EVENTUALLY CUT ALL ROAD DEATHS BY 20%



OBSERVED VEHICLE SPEEDS ABOVE THE SPEED LIMIT IN THE EU:

FROM	TO	ON
35%	TO 75%	ON URBAN ROADS
9%	TO 63%	ON RURAL ROADS
23%	TO 59%	ON MOTORWAYS

SOLUTIONS FOR SAFER SPEEDS:

 SAFE AND CREDIBLE SPEED LIMITS	 SELF-EXPLAINING, SELF-ENFORCING ROADS	 VEHICLES THAT HELP DRIVERS TO COMPLY WITH SPEED LIMITS
 STRICTER LAWS	 ROAD USER EDUCATION	 EFFECTIVE AND FREQUENT TRAFFIC LAW ENFORCEMENT ACTIVITIES

ETSC
European Transport Safety Council

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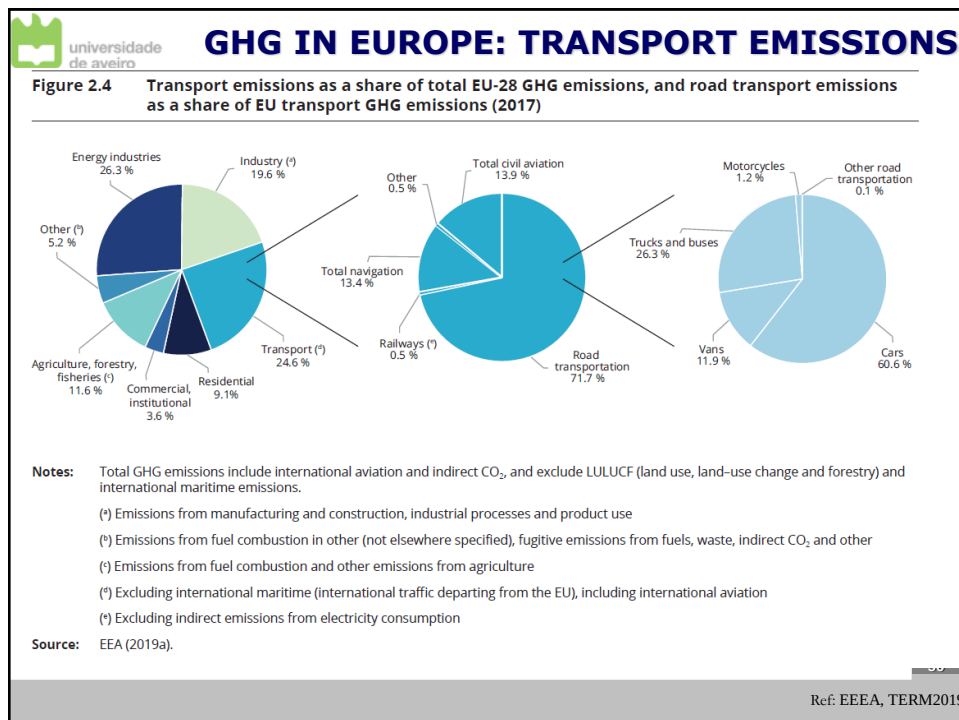
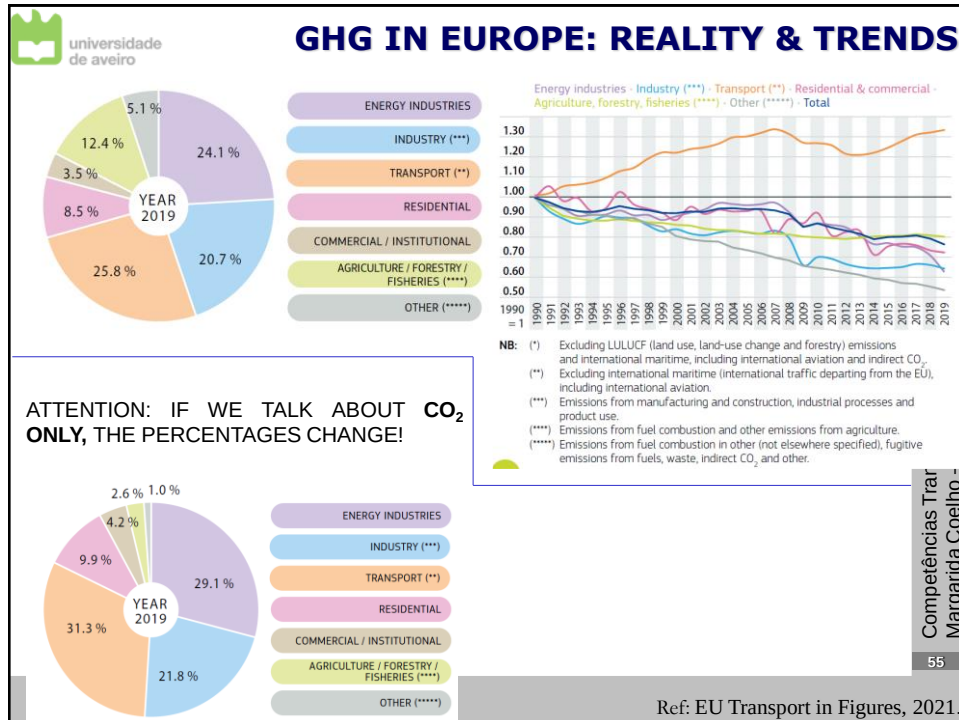
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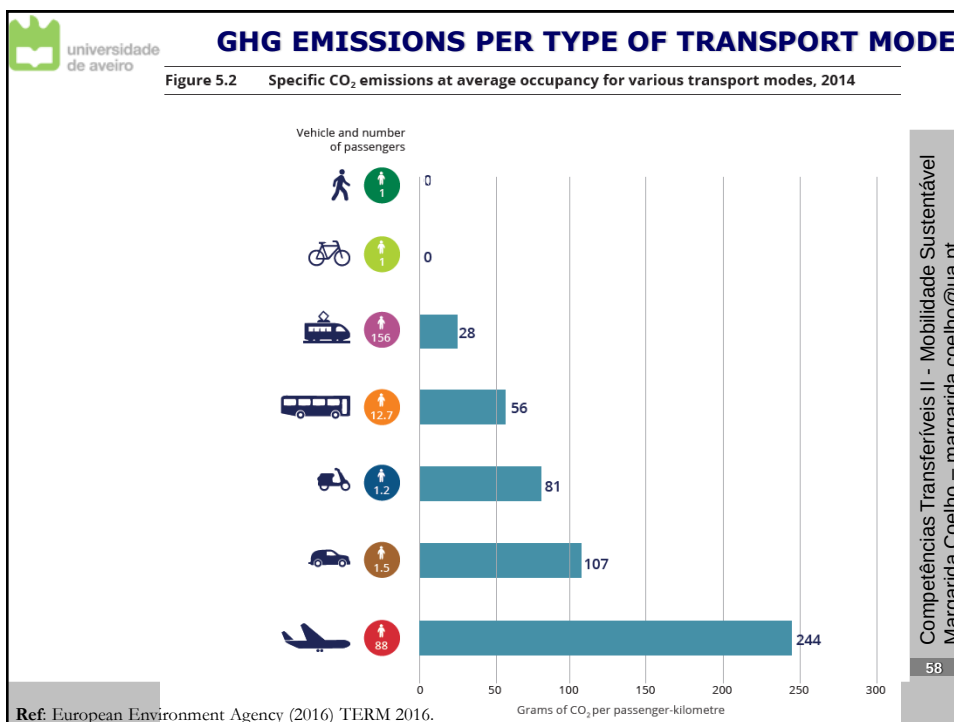
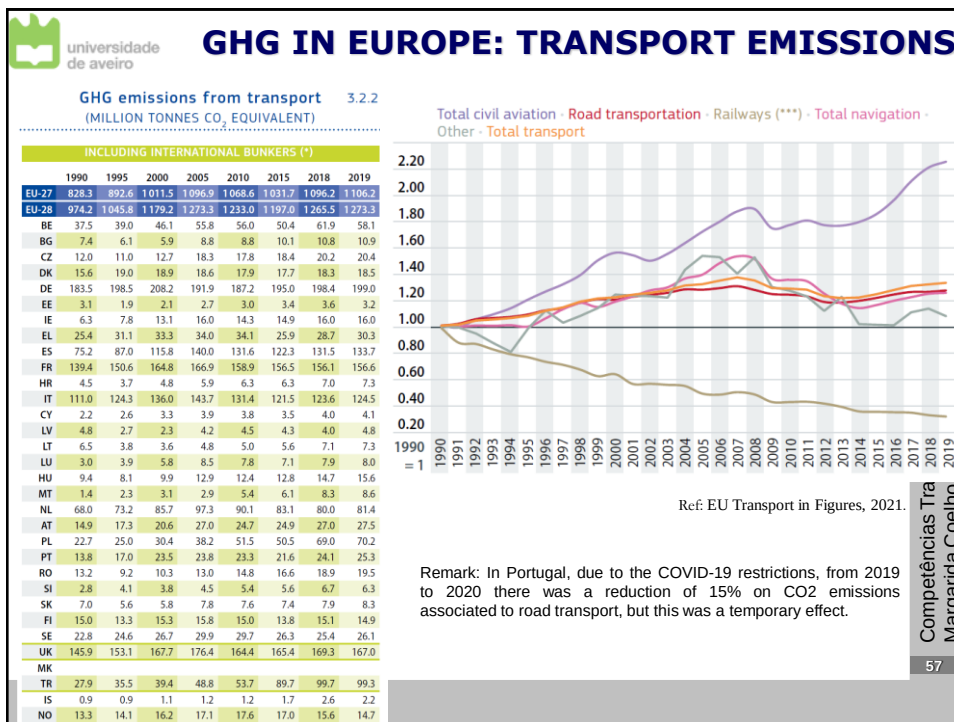
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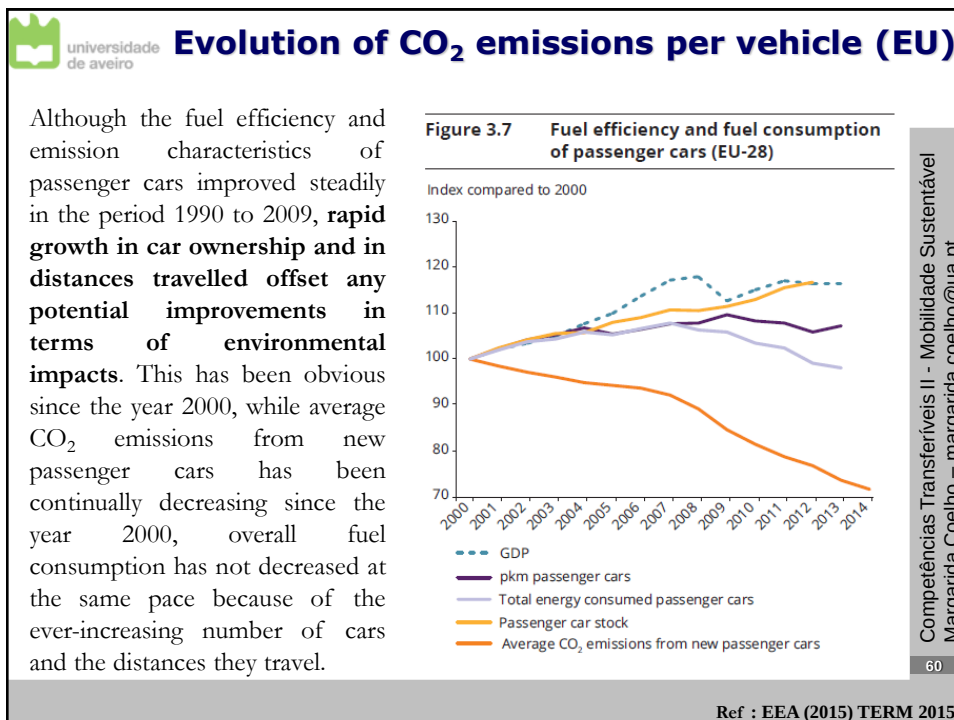
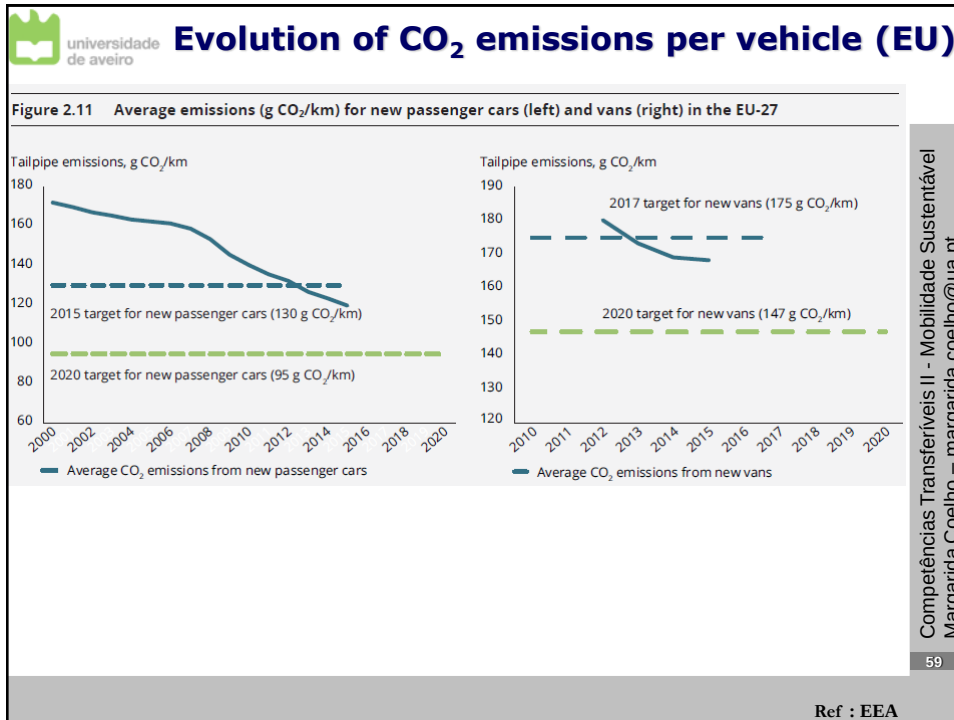
CHALLENGE #3 – ENERGY CONSUMPTION / EMISSIONS

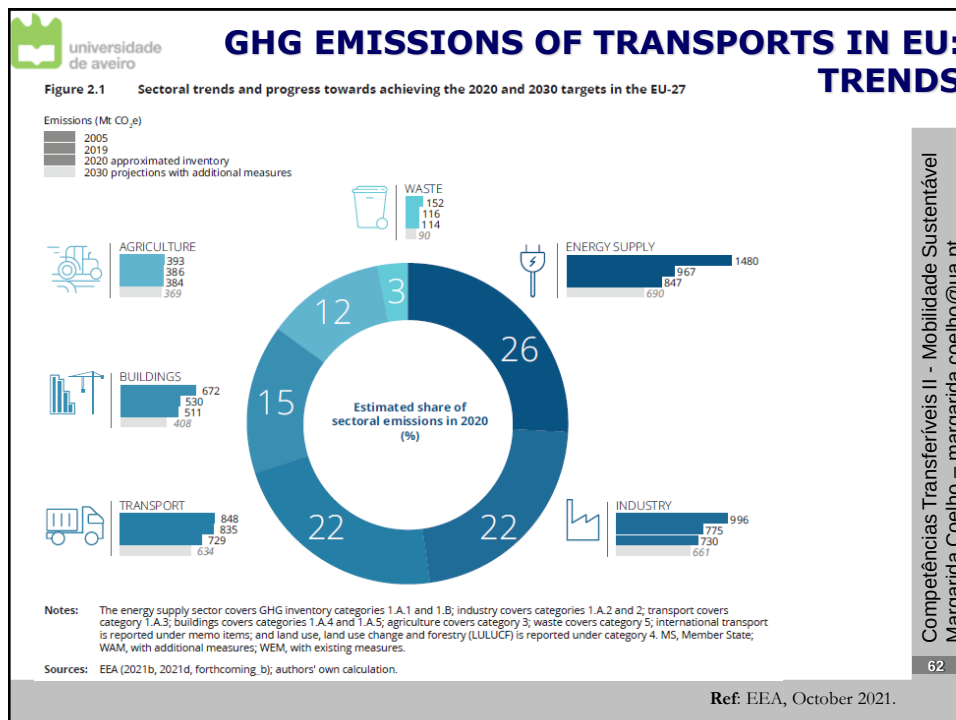
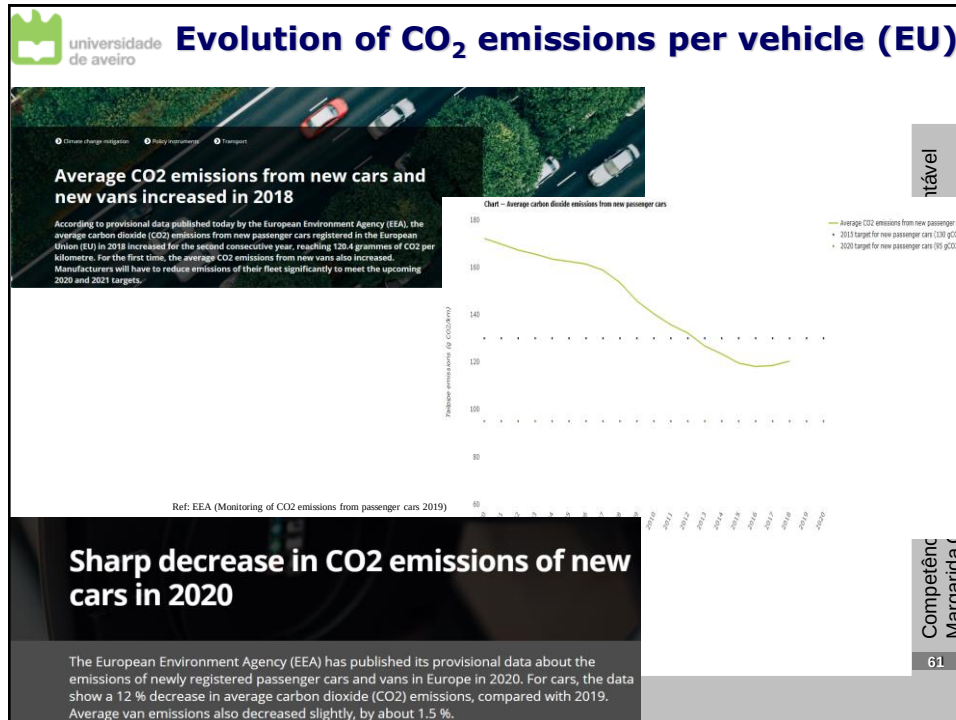
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GHG EMISSIONS OF TRANSPORTS IN EU: TARGETS

Contexto Regulatório

Existem *timings* concretos para a dimensão do CO₂, ao nível da União Europeia, com duas principais metas:

2020 *Green Deal*

2030 *Fit for 55*
Meta 1: Redução em 55% emissões até 2030

2035 *IPNEC 2030*

2050 *Green Deal*
Meta 2: Neutralidade carbónica até 2050

2023: Tributação dos produtos energéticos (combustíveis)
2026: Alargamento do CELE ao sector dos transportes rodoviário e marítimo

Reduzir emissões GEE em 45-55% e em 40% no setor dos transportes (vs. 2005)

47% energias renováveis no consumo bruto de energia e 20% no transporte e mobilidade

~ 40% geração energia renovável
~ 30% redução consumo energia final ~ Restrição normas de desempenho CO₂ - novos veículos ligeiros

~ Fase-out/curva de veículos a diesel. Ambição de novos veículos com 0 emissões

~ 90% redução de emissões no sector dos transportes
~ Restantes 10% serão compensados

Potencial aumento do custo dos combustíveis e eletricidade

Tendência zero emissões para a mobilidade

Renovation wave do parque de edifícios, incluindo armazéns

Fonte: Comissão Europeia, 2021; Pacote "Fit for 55", 2021; GRRP, 2021; Relatório para a Neutralidade Carbónica 2050, 2019.

Ref: KPMG, Observatório Automóvel, 2022.

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GHG EMISSIONS OF TRANSPORTS IN EU: TARGETS

Roteiro para a Neutralidade Carbónica 2050

Portugal criou o RNC2050 que define o compromisso e a estratégia do Governo Português para ir ao encontro do European *Green Deal* e às suas 2 metas chave

ROTEIRO PARA A NEUTRALIDADE CARBÓNICA (LONGO-PRAZO):

- Estabelece a visão de longo prazo, mas com ação imediata, em torno de três eixos interconectados: a valorização do território e do habitat, a economia circular e a decarbonização da sociedade e a transição energética
- O RNC2050, estabelece, de forma sustentada, a trajetória para atingir a neutralidade carbónica em 2050
- Atingir a neutralidade carbónica em Portugal implica reduzir as emissões de gases com efeito de estufa em mais de 85%, em relação a 2005, e garantir uma capacidade de sequestro agrícola e florestal de carbono na ordem dos 13 milhões de toneladas
- Em linha com os resultados do Relatório IPCC sobre 1,5 °C¹, é até 2030 que será colocado o maior esforço de redução de emissões (55% de redução das emissões de CO₂) e esta ambição já se traduziu no Plano Nacional Energia e Clima 2030, com uma clara aposta na transição energética e na mobilidade sustentável.

ROTEIRO PARA A NEUTRALIDADE CARBÓNICA 2050 (RNC2050)

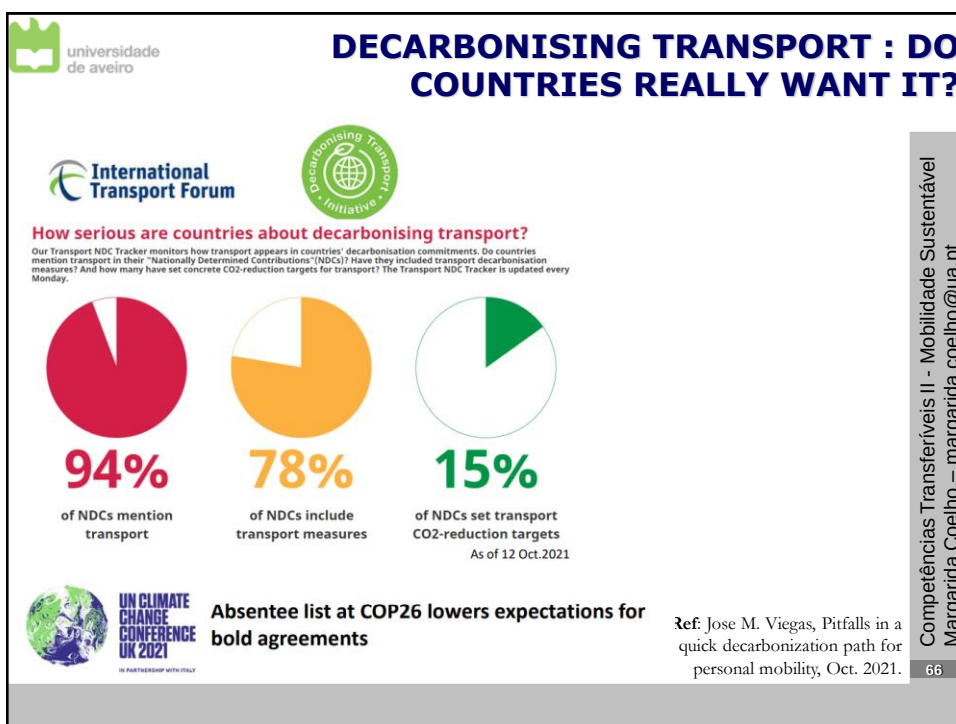
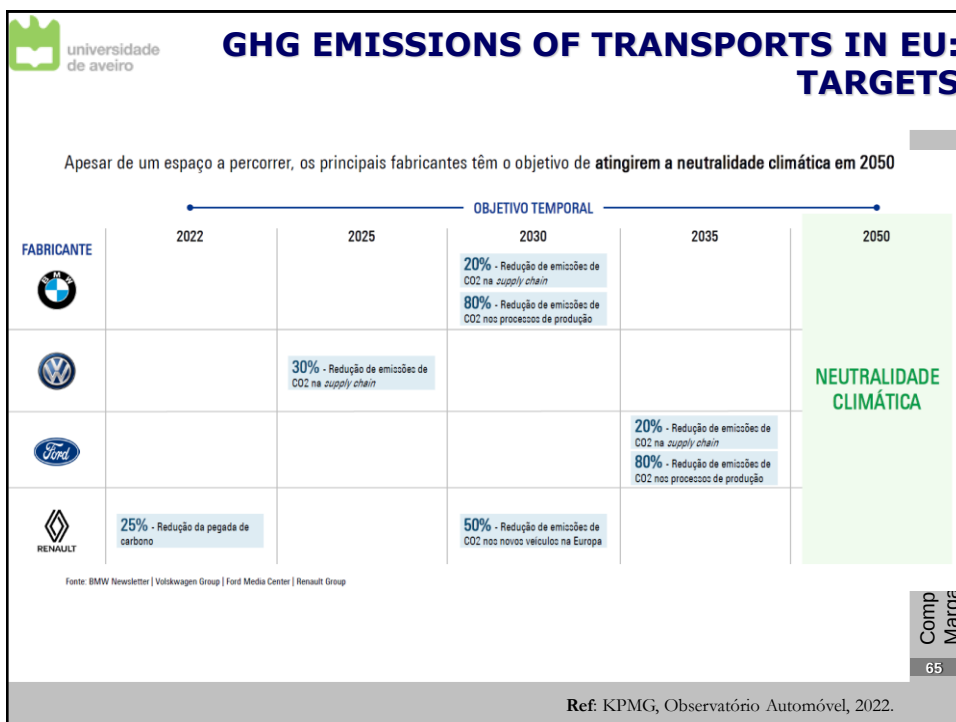
ESTRATÉGIA DE LONGO PRAZO PARA A NEUTRALIDADE CARBÓNICA DA ECONOMIA PORTUGUESA EM 2050

¹) O Painel Intergovernamental das Nações Unidas refere que para mitigar os riscos das alterações climáticas, é necessário reduzir o aumento de temperatura a nível global em 1,5 °C até 2030

Ref: KPMG, Observatório Automóvel, 2022.

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CO₂ IS IMPORTANT... BUT WHAT ABOUT THE OTHER POLLUTANTS?

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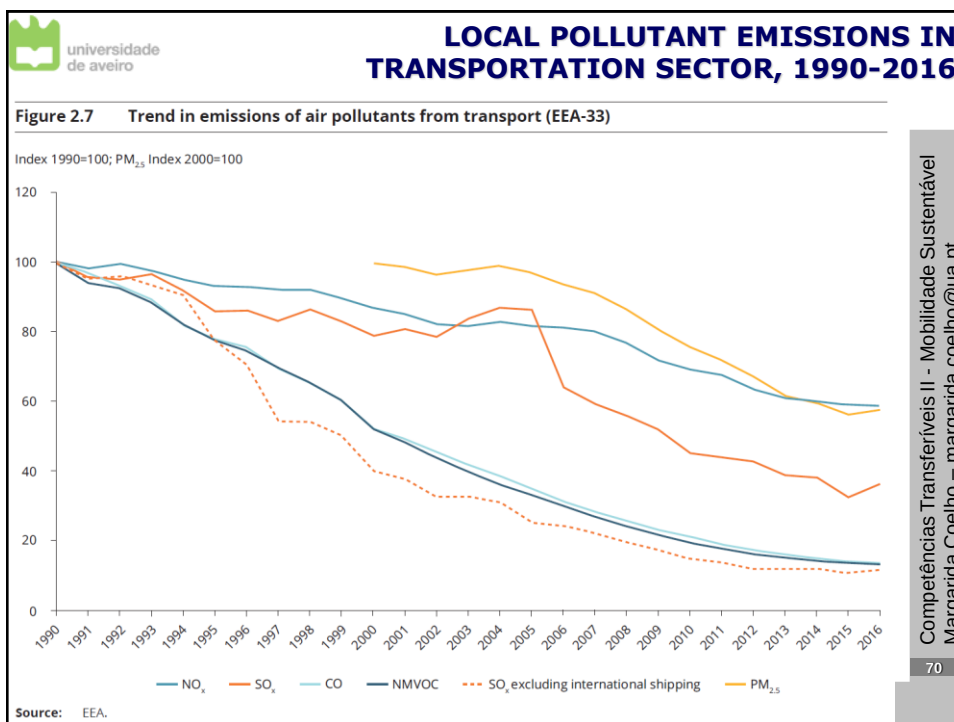
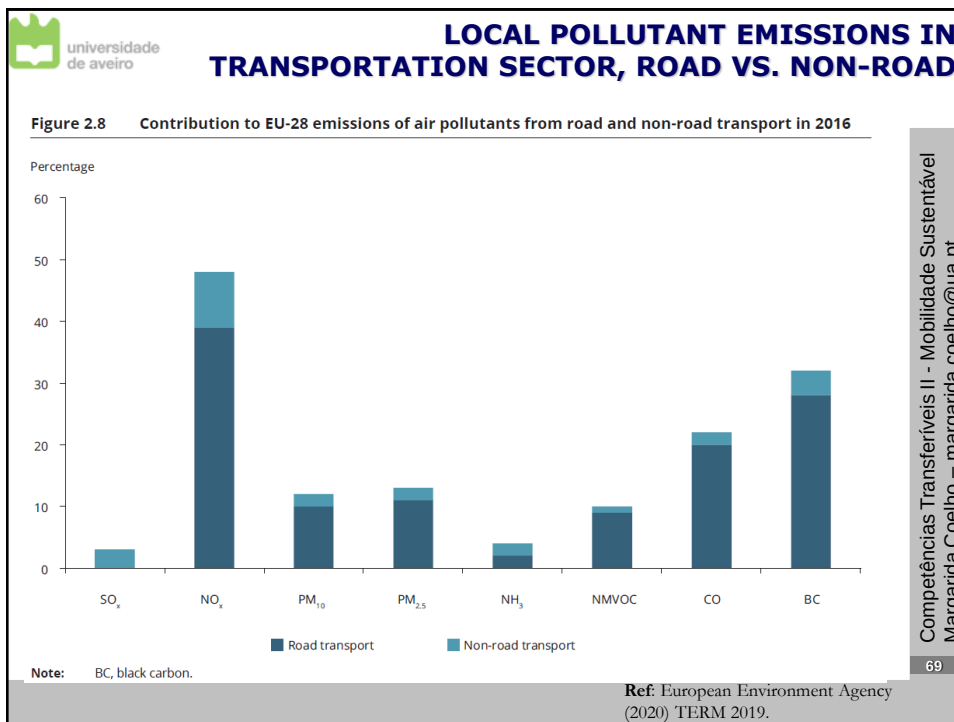
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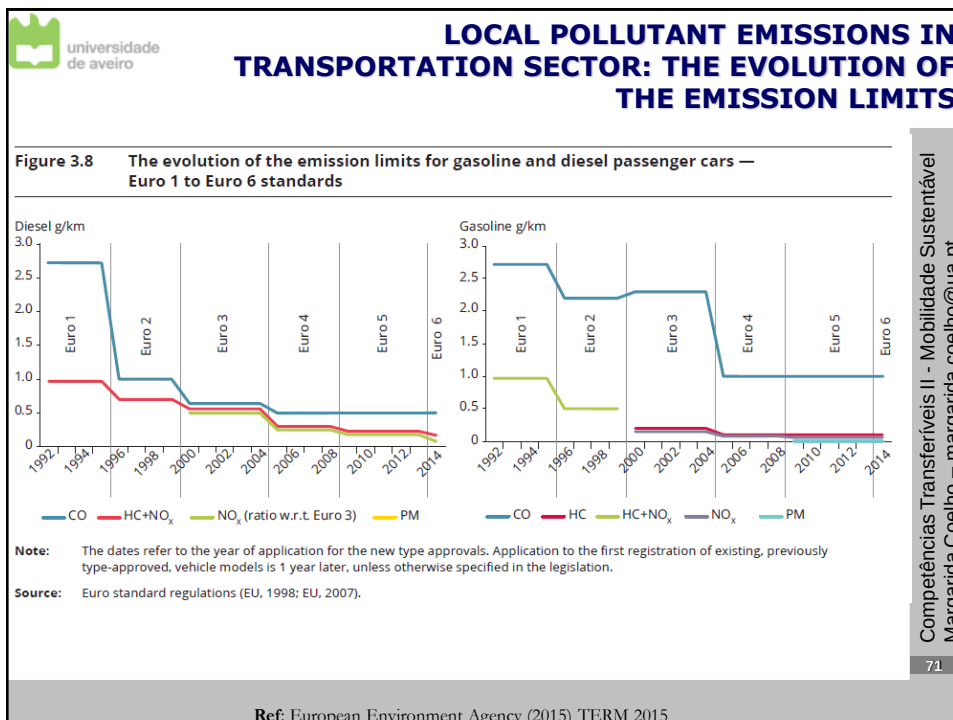
LOCAL POLLUTANT EMISSIONS IN TRANSPORTATION SECTOR: FACTS

- Despite considerable improvements in past decades, **air pollution is still responsible for more than 400 000 premature deaths in Europe each year.**
- It also continues to damage vegetation and ecosystems.
- Significant progress has been made since 1990 in reducing the emissions of many air pollutants from the transport sector in Europe. All transport modes have decreased their emissions, except for international aviation and shipping, for which emissions of each pollutant have increased since 1990.

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EMISSION MEASUREMENTS / EMISSION STANDARDS

The European emission standards define the limits for emissions per vehicle

Table 1
EU Emission Standards for Passenger Cars (Category M₁⁴)

Stage	Date	CO	HC	HC+NOx	NOx	PM	PN
		g/km					#/km
Compression Ignition (Diesel)							
Euro 1†	1992.07	2.72 (3.16)	-	0.97 (1.13)	-	0.14 (0.18)	-
Euro 2, IDI	1996.01	1.0	-	0.7	-	0.08	-
Euro 2, DI	1996.01 ^a	1.0	-	0.9	-	0.10	-
Euro 3	2000.01	0.64	-	0.56	0.50	0.05	-
Euro 4	2005.01	0.50	-	0.30	0.25	0.025	-
Euro 5a	2009.09 ^b	0.50	-	0.23	0.18	0.005 ^f	-
Euro 5b	2011.09 ^c	0.50	-	0.23	0.18	0.005 ^f	6.0×10 ¹¹
Euro 6	2014.09	0.50	-	0.17	0.08	0.005 ^f	6.0×10 ¹¹
Positive Ignition (Gasoline)							
Euro 1†	1992.07	2.72 (3.16)	-	0.97 (1.13)	-	-	-
Euro 2	1996.01	2.2	-	0.5	-	-	-
Euro 3	2000.01	2.30	0.20	-	0.15	-	-
Euro 4	2005.01	1.0	0.10	-	0.08	-	-
Euro 5	2009.09 ^b	1.0	0.10 ^d	-	0.06	0.005 ^{e,f}	-
Euro 6	2014.09	1.0	0.10 ^d	-	0.06	0.005 ^{e,f}	6.0×10 ¹¹ e.g

⁴ At the Euro 1..4 stages, passenger vehicles > 2,500 kg were type approved as Category N₁ vehicles

† Values in brackets are conformity of production (COP) limits

a. until 1999.09.30 (after that date DI engines must meet the IDI limits)

b. 2011.01 for all models

c. 2013.01 for all models

d. and NMHC = 0.068 g/km

e. applicable only to vehicles using DI engines

f. 0.0045 g/km using the PMP measurement procedure

g. 6.0×10¹² 1/km within first three years from Euro 6 effective dates

Ref: <http://www.dieselnit.com/standards/eu/ld.php>

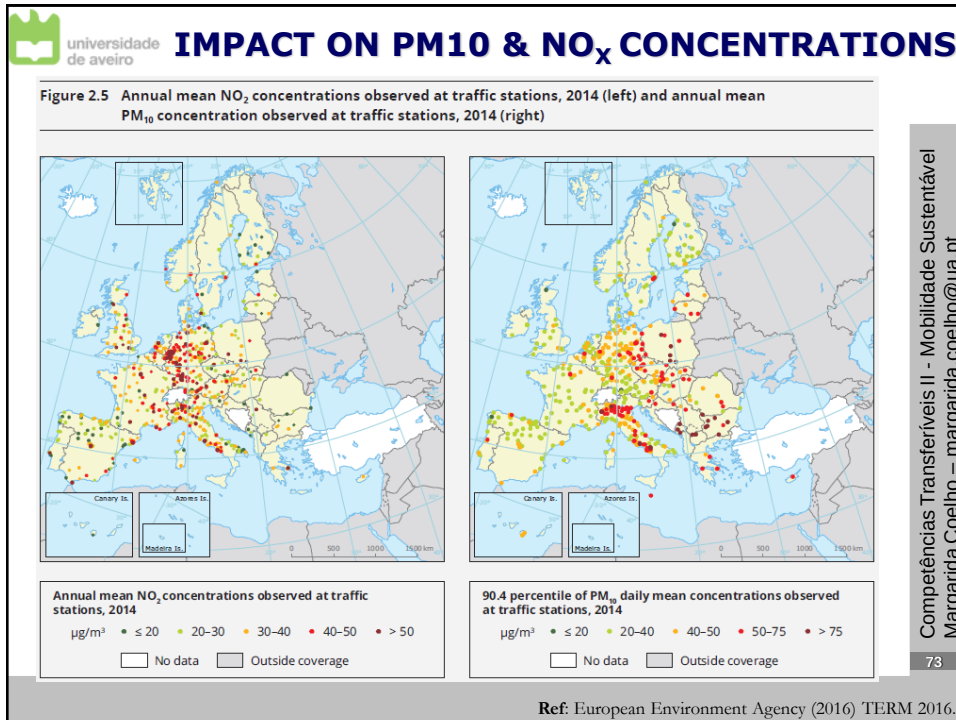
Objective: EURO 7/VII already in 2025! The auto industry asks extra time.

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E-LEARNING:

EUROPEAN STRATEGY FOR SMART AND SUSTAINABLE MOBILITY

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CONCLUSIONS & DISCUSSION

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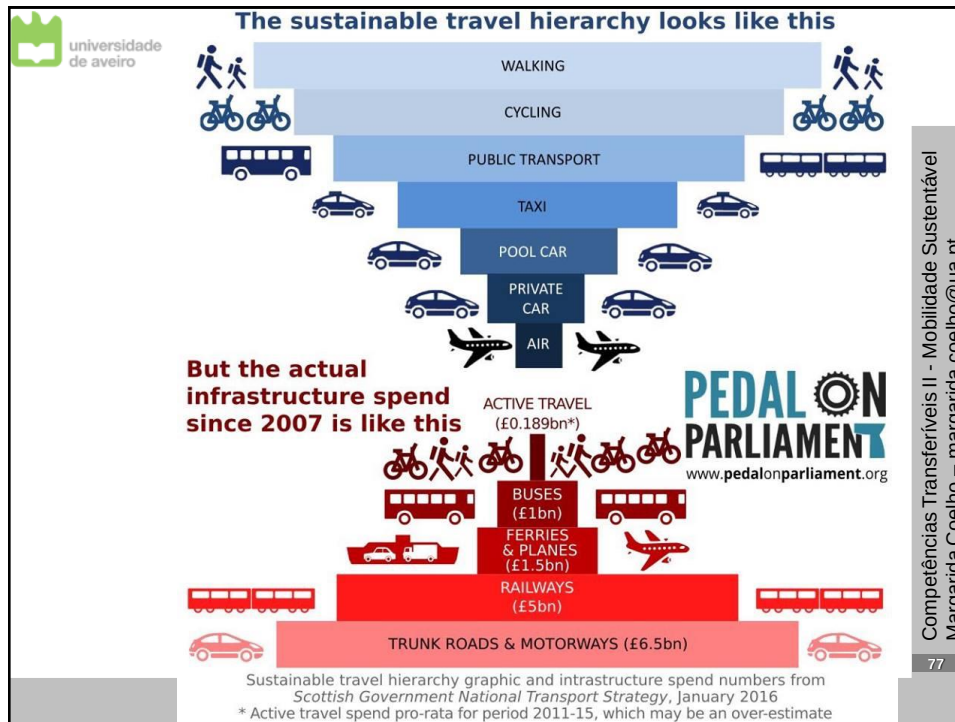
TRANSPORTATION: ONE OF THE MOST DIFFICULT SECTORS TO CONTROL



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DECARBONISING TRANSPORT : MAIN CHALLENGES FOR PERSONAL MOBILITY

Mainstream of politicians looking at reduction of GHG emissions via:

- Electric vehicles & (fast) Charging points / stations
- High hopes on new battery technologies
- Modal shift to scheduled public transport
- In some countries, also e-bikes

But (because there is always a but...):

- Great diversity of personal situations in society, many of which with a difficult fit in the “new mobility”, due to:
 - Affordability to shift to EV
 - Flexibility / Spatial coverage of alternative options to private car
- Mobility options are a key factor of many people’s professional and personal frameworks

⇒ **Risk of low acceptance / resistance**

(Adapted from Jose M. Viegas, *Pitfalls in a quick decarbonization path for personal mobility*, Oct. 2021.)

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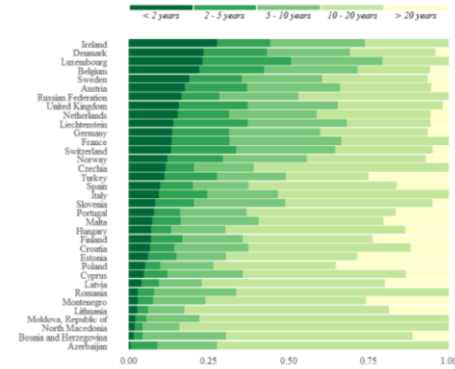
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DECARBONISING TRANSPORT : MAIN CHALLENGES FOR PERSONAL MOBILITY

Dealing with the senior car fleet:

- Many countries with 40%-60% vehicles above 10 years
- These old vehicles are high emitters
- Many of these cars critical for daily life, especially in peri-urban and rural areas, owned by people with low disposable income
- Access to jobs and professional routes, often with high daily distance driven
- Modal shift virtually impossible
- Trade-in values very low, replacement by EV unaffordable with current subsidy levels
- Low level of tolerance to behaviour-change

Share of passenger car fleet by vehicle age category and country, 2017 or most recent year



(Adapted from Jose M. Viegas, Pitfalls in a quick decarbonization path for personal mobility, Oct. 2021.)

DECARBONISING TRANSPORT : MAIN CHALLENGES FOR PERSONAL MOBILITY

Great difficulties lie ahead for the transition:

- Transition will be turbulent and difficult
- Societies are complex: big diversity of situations and interactions between the different segments
- Many jobs (and interests in general) in industries and services connected to "mobility as usual"
- Many lifestyles with very difficult fit to scheduled public transport
- Core of the problem is very similar all over the EU, but with some diversity of parameters and even some different features associated with land-use, social habits and purchasing power

Decarbonization is the priority and seems technologically feasible in personal mobility

- But the other objectives of Sustainable Mobility must not be forgotten
- In particular, universal equitable access may be at risk for some segments
- Solutions excessively focused on technological progress and aiming at "maximum utility" or "best option on average" will face serious acceptance issues
- A range of solutions must be available to each citizen (?)
- Some solutions may require incentives / nudges to increase and accelerate their acceptance

(Adapted from Jose M. Viegas, Pitfalls in a quick decarbonization path for personal mobility, Oct. 2021.)

